BK166X Series Interface Description

UG01-BK166X-E10 V1.9.28

2024/11/13

Contents

1. About This Document

This document describes the interface of the BK166X series standard precision GNSS firmware. This document includes the following sections:

- NMEA Protocol, which introduces the NMEA plaintext protocol.
- BK Protocol, which introduces the BK binary output and input protocol.
- RTCM Protocol, which introduces the RTCM protocol.
- Commonly Used Commands, which provides commonly used configuration commands.
- AGNSS, which introduces the AGNSS workflow.

2. NMEA Protocol

2.1 NMEA Frame Structure

NMEA messages sent by the GNSS receiver are based on NMEA 0183 Version 4.10 standard. Figure 2-1 shows the structure of a NMEA protocol message (called "sentences" in the standard).

Figure 2-1 NMEA Frame Structure

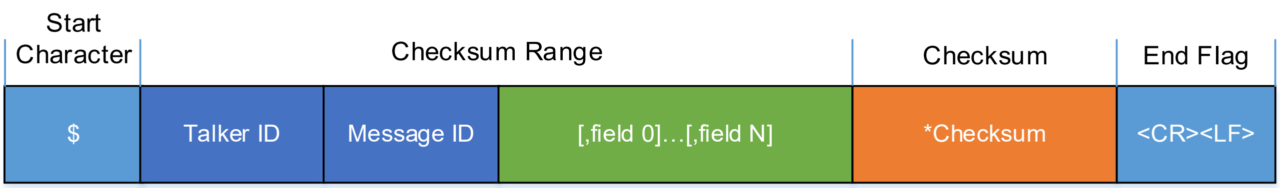


Table 2-1 NMEA Frame Structure

Field	Description
$	Start character, always ‘$’
Talker ID	Satellite system type. Two characters, only digits and uppercase letters, including “GP” (GPS/QZSS/SBAS), “GN” (GNSS), “BD” (Beidou), “GL” (GLONASS), “IR” (IRNSS), “GA” (Galileo). Note: Talker ID “PO” is used for BK166X proprietary information messages.
Message ID	Sentence identifier. Three characters, only digits and uppercase letters, including “GGA”, “GSA”, “GSV”, “RMC”, etc. Note: “CNR”, “CLK”, “LRS”, “ANT”, “JAM”, and “INS” are specialized messages for BK166X.
Data Field (Payload)	The data fields contain the actual information being transmitted. Each field is separated by a comma (','). The number and order of the fields depend on the specific sentence type.
Checksum	Always begins with a ‘*’, followed by two hex characters. The checksum is exclusive OR between ‘$’ and ‘*’.
<CR><LF>	End flag, always <CR><LF>

2.2 Standard Messages

The currently supported standard NMEA messages defined by the NMEA standard are shown in Table 2-2.

Table 2-2 Standard NMEA Messages

Message ID	Description
GGA	Global Positioning System Fix Data: time, position, and fix related data provided by the GNSS receiver
GSA	GNSS DOP and Active Satellites: GNSS receiver operating mode, satellites used in the navigation solution reported by the GGA sentence and DOP values
GSV	GNSS Satellites in View: number of satellites (SV) in view, satellite ID numbers, elevation, azimuth, and C/N0
RMC	Recommended Minimum Specific GNSS Data: time, date, position, course, and speed data provided by the GNSS receiver
THS	True Heading and Status: heading and positioning mode provided by the GNSS receiver
GLL	Geographic Position – Latitude/Longitude, with time of position fix and status
GST	GNSS Pseudorange Error Statistics
GMP	GNSS Map Projection Fix Data
ZDA	Time and date provided by GNSS receiver
DTM	Outputs the offset between the Local and the Reference.
GBS	Output the results of the Autonomous Integrity Detection Algorithm (RAIM)
GNS	Outputs Time, Position, and Information about GNSS positioning.

2.2.1 NMEA_GGA

Table 2-3 NMEA_GGA

Message	NMEA_GGA
NMEA Message ID	GGA
Description	Time, position, and fix related data provided by the GNSS receiver
Firmware	-
Comment	-
Information	Number of fields: 15
Structure	$XXGGA,UTC,LAT,DIR_LAT,LON,DIR_LONG,QUAL_IND,NUM_SV,HDOP,ORTH,ORTH_UNIT,GEOID,GEOID_UNIT,AGE,REFS_ID*cs<CR><LF>

© 2022-2024 Beken Corporation

Example	$GNGGA,093314.00,3110.4880379,N,12135.9872231,E,1,37,0.5,17.362,M,0.000,M,,*74<CR><LF>

Table 2-4 Payload of NMEA_GGA Message

Field No.	Name	Unit	Format	Example	Description
0	$XXGGA	-	string	$GNGGA	Talker ID: GN for GNSS Message ID: GGA
1	UTC	-	HHMMSS.ss	093314.00	UTC time
2	LAT	-	DDMM.mmmmm	3110.4880379	Latitude
3	DIR_LAT	-	character	N	Direction of latitude: • N: North • S: South
4	LON	-	DDDMM.mmmmm	12135.9872231	Longitude
5	DIR_LONG	-	character	E	Direction of longitude: • E: East • W: West
6	QUAL_IND	-	digit	1	Fix quality indicator: • 0: No fix • 1: GPS fix • 2: Differential GPS fix (DGNSS, SBAS) • 3: PPP fix • 4: RTK fix • 5: RTK float fix • 6: INS dead reckoning fix • 7:Base
7	NUM_SV	-	numeric	37	Number of satellites used
8	HDOP	-	numeric	0.5	Horizontal Dilution of Precision
9	ORTH	m	numeric	17.362	Orthometric height
10	ORTH_UNIT	-	character	M	Orthometric height units: M (meters)
11	GEOID	m	numeric	-	Geoid separation
12	GEOID_UNIT	-	character	M	Geoid separation units: M (meters)
13	AGE	s	numeric	-	Age of differential corrections

Field No.	Name	Unit	Format	Example	Description
14	REFS_ID	-	numeric	-	ID of station providing differential corrections
15	cs	-	-	*74	Checksum
16	<CR><LF>	-	character	-	Carriage return and line feed

2.2.2 NMEA_GSA

Table 2-5 NMEA_GSA

Message	NMEA_GSA
NMEA Message ID	GSA
Description	GNSS receiver operating mode, satellites used in the navigation solution reported by the GGA sentence, and DOP values.
Firmware	-
Comment	-
Information	Number of fields: 10
Structure	$XXGSA,OP_MODE,FIX_MODE[,PRN],PDOP,HDOP,VDOP,SYS_ID*cs<CR><LF>
Example	$GPGSA,A,3,01,07,08,14,17,21,27,30,194,195,199,,0.9,0.5,0.7,1*1C<CR><LF>

Table 2-6 Payload of NMEA_GSA Message

Field No.	Name	Unit	Format	Example	Description
0	$XXGSA	-	string	$GPGSA	Talker ID: GP for GPS Message ID: GSA
1	OP_MODE	-	character	A	Operation mode: • M = Manual • A = Automatic
2	FIX_MODE	-	digit	3	Fix mode: • 1 = Not available • 2 = 2D • 3 = 3D
3	[,PRN]	-	numeric	01	PRN numbers of Satellites, see Table 2-7.

Field No.	Name	Unit	Format	Example	Description
					This field can be repeated up to 12 times.
4	PDOP	-	numeric	0.9	Position dilution of precision
5	HDOP	-	numeric	0.5	Horizontal dilution of precision
6	VDOP	-	numeric	0.7	Vertical dilution of precision
7	SYS_ID	-	numeric	37	System ID, see Table 2-7.
8	cs	-	-	*1C	Checksum
9	<CR><LF>	-	character	-	Carriage return and line feed

Table 2-7 System and PRN Definition

System Name	Talker ID (System ID)	PRN Number	Signal	
			Signal ID	Signal Channel
GPS	GP(1)	1–32: GPS 33–64: SBAS 65–99: Reserved	0	All Signals
			1	L1 C/A
			2	L1 P(Y)
			3	L1 M
			4	L2 P(Y)
			5	L2C M
			6	L2C L
			7	L5 I
			8	L5 Q
GLONASS	GL(2)	33–64: SBAS 65–99: GLONASS	0	All Signals
			1	G1 C/A
			2	G1 P
			3	G2 C/A
			4	G2 P
Galileo	GA(3)	1–36: GAL 37–64: GAL-SBAS 65–99: Reserved	0	All Signals
			1	E5 A
			2	E5 B
			3	E5 A+B
			4	E6 A
			5	E6 B

System Name	Talker ID (System ID)	PRN Number	Signal	
			Signal ID	Signal Channel
			6 7	L1 A L1 B
Beidou	BD(4)	1–63: Beidou 64–99: Reserved	0 1 2 3 5 6 7	All Signals B1 I B1 C B2 A B2 I / B2 B B2 A+B B3 I
IRNSS	IR(5)	1–14: IRNSS 15–64: SBAS	0 1	All Signals L5 I
QZSS	GP(1)	193–202: QZSS	0 1 7	All Signals L1 C/A L5 I

2.2.3 NMEA_GSV

Table 2-8 NMEA_GSV

Message	NMEA_GSV
NMEA Message ID	GSV
Description	Number of satellites (SV) in view, satellite ID numbers, elevation, azimuth, signal strength (C/N0), and signal ID
Firmware	-
Comment	-
Information	Number of fields: 10
Structure	$XXGSV,MSG_NUM,MSG_ID,SV_NUM[,PRN,EL,AZ,CN0],SIG*cs<CR><LF>
Example	$GPGSV,5,1,17,01,73,173,42,07,57,238,45,08,39,046,42,14,25,313,36,1*6D<CR><LF> $GPGSV,5,2,17,17,17,264,25,21,66,056,42,27,10,063,28,28,00,000,29,1*6E<CR><LF> $GPGSV,5,3,17,30,48,284,40,194,66,121,40,195,76,068,40,196,00,000,14,1*56<CR><LF> $GPGSV,5,4,17,199,53,170,15,56,00,000,43,57,00,000,32,03,09,158,00,1*57<CR><LF> $GPGSV,5,5,17,193,05,155,00,,,,,,,,,,,,1*6D<CR><LF>

Table 2-9 Payload of NMEA_GSV

Field No.	Name	Unit	Format	Example	Description
0	$XXGSV	-	string	$GPGSV	• Talker ID: GP for GPS • Message ID: GSV
1	MSG_NUM	-	digit	5	Total number of messages of this type in this epoch
2	MSG_ID	-	digit	1	The index of current message in total number of GSV messages
3	SV_NUM	-	digit	17	Number of visible satellites
4	[,PRN]	-	numeric	01	PRN numbers of Satellites, see Table 2-7.
5	[,EL]	deg	numeric	73	Elevation in degrees (≤90)
6	[,AZ]	deg	numeric	173	Azimuth from True North in degrees (range: 0–359)
7	[,CN0]	dBHz	numeric	42	Signal strength C/N0 (range: 0–99). Fields No.4–7 can be repeated for 1 to 4 times.
8	SIG_ID	-	digit	1	NMEA-defined GNSS signal ID, See Table 2-7 (only available in NMEA 4.10 and later)
9	cs	-	hexadecimal	*6D	Checksum
10	<CR><LF>	-	character	-	Carriage return and line feed

2.2.4 NMEA_VTG

Table 2-10 NMEA_VTG

Message	NMEA_VTG
NMEA Message ID	VTG
Description	Course over ground and ground speed
Firmware	-
Comment	-
Information	Number of fields: 12
Structure	$XXVTG,COGT,T,COGM,M,KNOTS,N,KPH,K,POS_MODE*cs<CR><LF>

Example	$GNVTG,50.55,T,,M,33.27,N,61.62,K,A*15

Table 2-11 Payload of NMEA_VTG

Field No.	Name	Unit	Format	Example	Description
0	$XXVTG	-	string	$GNVTG	• Talker ID: GN for GNSS • Message ID: VTG
1	COGT	deg	numeric	50.55	Course over ground (true) (range: 0–360)
2	T	-	character	T	Course over ground units: T (degrees true, fixed field)
3	COGM	deg	numeric	-	Course over ground (magnetic), not supported.
4	M	-	character	M	Course over ground units: M (degrees magnetic, fixed field)
5	KNOTS	knots	numeric	33.27	Speed over ground
6	N	-	character	N	Speed over ground units: N (knots, fixed field)
7	KPH	km/h	numeric	61.62	Speed over ground
8	K	-	character	K	Speed over ground units: K (kilometers per hour, fixed field)
9	POS_MODE	-	character	A	Mode indicator, see Table 2-14.
10	cs	-	hexadecimal	15	Checksum
11	<CR><LF>	-	character	-	Carriage return and line feed

2.2.5 NMEA_RMC

Table 2-12 NMEA_RMC

Message	NMEA_RMC
NMEA Message ID	RMC
Description	Time, date, position, course, and speed data provided by the GNSS receiver
Firmware	-

Comment	-
Information	Number of fields: 16
Structure	$XXRMC,UTC,STATUS,LAT,DIR_LAT,LON,DIR_LONG,SPEED,COURSE,DATE,MAG,DIR_MAG,POS_MODE,NAV_ST*cs<CR><LF>
Example	$GNRMC,093314.00,A,3110.4880379,N,12135.9872231,E,3.09,30.61,090222,,,A,V*09<CR><LF>

Table 2-13 Payload of NMEA_RMC

Field No.	Name	Unit	Format	Example	Description
0	$XXRMC	-	string	$GNRMC	Talker ID: GN for GNSS Message ID: RMC
1	UTC	-	HHMMSS.ss	093314.00	UTC time of this epoch
2	STATUS	-	character	A	Status of position fix solution. See Table 2-14.
3	LAT	-	DDMM.mmmmm	3110.4880379	Latitude
4	DIR_LAT	-	character	N	Direction of latitude: • N: North • S: South
5	LON	-	DDDMM.mmmmm	12135.9872231	Longitude
6	DIR_LONG	-	character	E	Direction of longitude: • E: East • W: West
7	SPEED	knots	numeric	3.09	Speed over ground
8	COURSE	deg	numeric	30.61	Course over ground (range: 0–360)
9	DATE	-	ddmmyy	090222	UTC date in day, month, year format
10	MAG	deg	numeric	-	Magnetic variation value
11	DIR_MAG	-	character	-	Direction of Magnetic variation: • E: East • W: West
12	POS_MODE	-	character	A	Position fix mode, see Table 2-14.

Field No.	Name	Unit	Format	Example	Description
13	NAV_ST	-	character	V	Navigational status indicator: V
14	cs	-	hexadecimal	*09	Checksum
15	<CR><LF>	-	character	-	Carriage return and line feed

Table 2-14 Position Fix Flags

Position Fix Flag	STATUS[a] in $XXRMC	POS_MODE[b] in $XXRMC
No fix	V	N
GNSS fix	A	A
RTD or SBAS fix	A	D
RTK float fix	A	F
RTK fix	A	R
PPP	A	P
Reserved	A	E

a. Possible *STATUS* values: V = data invalid, A = data valid

b. Possible *POS_MODE* values: N = No fix, E = estimated (dead reckoning) fix, A = autonomous GNSS fix, D = differential GNSS fix, F = RTK float, R = RTK fixed, P = PPS fix

2.2.6 NMEA_THS

Table 2-15 NMEA_THS

Message	NMEA_THS
NMEA Message ID	THS
Description	True heading and status
Firmware	DUALANT mode
Comment	-
Information	Number of fields: 3
Structure	$XXTHS,HD,MODE*cs<CR><LF>
Example	$GNTHS,227.1,A*06<CR><LF>

Table 2-16 Payload of NMEA_THS

Field No.	Name	Unit	Format	Example	Description
0	$XXTHS	-	string	$GNTHS	Talker ID: GN for GNSS Message ID: THS
1	HD	deg	numeric	227.1	Heading of vehicle (true) (range: 0–360)
2	MODE	-	character	A	Mode indicator (cannot be null): • A = Autonomous mode • E = Estimated (dead reckoning) mode • M = Manual input mode • S = Simulator mode • V = Data not valid
3	cs	-	hexadecimal	*06	Checksum
4	<CR><LF>	-	character	-	Carriage return and line feed

2.2.7 NMEA_GLL

Table 2-17 NMEA_GLL

Message	NMEA_GLL
NMEA Message ID	GLL
Description	Latitude and longitude, with time of position fix and status
Firmware	-
Comment	-
Information	Number of fields: 10
Structure	$XXGLL,LAT,DIR_LAT,LON,DIR_LONG,UTC,NAV_ST,POS_MODE*cs<CR><LF>
Example	$GPGLL,3110.4880379,N,00833.91565,E,093314.00,A,A*60\r\n

Table 2-18 Payload of NMEA_GLL

Field No.	Name	Unit	Format	Example	Description
0	$XXGLL	-	string	$GNGLL	Talker ID: GN for GNSS Message ID: GLL

Field No.	Name	Unit	Format	Example	Description
1	LAT	-	DDMM.mmmmm	3110.4880379	Latitude
2	DIR_LAT	-	character	N	Direction of latitude: • N: North • S: South
3	LON	-	DDMM.mmmmm	12135.9872231	Longitude
4	DIR_LONG	-	character	E	Direction of longitude: • E: East • W: West
5	UTC	-	HHMMSS.ss	093314.00	UTC time of this epoch
6	NAV_ST	-	character	A	Navigational status indicator
7	POS_MODE	-	character	A	Position fix mode, see Table 2-14.
8	cs	-	hexadecimal	*60	Checksum
9	<CR><LF>	-	character	-	Carriage return and line feed

2.2.8 NMEA_GST

Table 2-19 NMEA_GST

Message	NMEA_GST
NMEA Message ID	GST
Description	GNSS pseudorange error statistics
Firmware	-
Comment	-
Information	Number of fields: 11
Structure	$xxGST,UTC,RMS_RANGE,STD_MAJOR,STD_MINOR,ORIENT,STD_LAT,STD_LONG,STD_ALT*cs
Example	$GNGST,133144.000,4.30,2.00,1.90,133.6000,1.9000,1.9000,8.2000*49

Table 2-20 Payload of NMEA_GST

Field No.	Name	Unit	Format	Example	Description
0	$XXGST	-	string	$GNGST	• Talker ID: GN for GNSS

Field No.	Name	Unit	Format	Example	Description
					• Message ID: GST
1	UTC	-	HHMMSS.ss	133144.000	UTC time
2	RMS_RANGE	m	numeric	4.30	RMS value of the standard deviation of the ranges (not support)
3	STD_MAJOR	m	numeric	2.00	Standard deviation of semi-major axis (not supported)
4	STD_MINOR	m	numeric	1.90	Standard deviation of semi-minor axis (not supported)
5	ORIENT	deg	numeric	133.6000	Orientation of semi-major axis (not supported)
6	STD_LAT	m	numeric	1.9000	Standard deviation of latitude error
7	STD_LONG	m	numeric	1.9000	Standard deviation of longitude error
8	STD_ALT	m	numeric	8.2000	Standard deviation of altitude error
9	cs	-	hexadecimal	*49	Checksum
10	<CR><LF>	-	character	-	Carriage return and line feed

2.2.9 NMEA_GMP

Table 2-21 NMEA_GMP

Message	NMEA_GMP
NMEA Message ID	GMP
Description	GNSS map projection fix data
Firmware	-
Comment	-
Information	Number of fields: 17
Structure	$xxGMP,UTC,MAP_ID,MAP_ZONE,NORTH,EAST,POS_MODE,NUM_SV,HDOP,ORTH,GEOID,AGE,REFS_ID,NAV_IND*cs<CR><LF>
Example	$GNGMP,065119.00,UTM,M51,3453206.069,366945.770,A,33,0.6,27.659,0.000,,,*39

© 2022-2024 Beken Corporation

Table 2-22 Payload of NMEA_GMP

Field No.	Name	Unit	Format	Example	Description
0	$XXGMP	-	string	$GNGMP	Talker ID: GN for GNSS Message ID: GMP
1	UTC	-	HHMMSS.ss	133144.000	UTC time of this epoch
2	MAP_ID	-	string	UTM	Projection coordinate system, only supports UTM projection
3	MAP_ZONE	-	MXX	M51	Projection zone number
4	NORTH	m	numeric	3453206.069	North component of projection coordinates In UTM projection, to avoid negative parameters: NORTH = NORTH (Northern Hemisphere) NROTH = NORTH + 10 000 000 (Southern Hemisphere)
5	EAST	m	numeric	366945.770	East component of projection coordinates In UTM projection, to avoid negative parameters: EAST = EAST + 500 000
6	POS_MODE	-	string	A	Position fix mode, see Table 2-14
7	NUM_SV	-	numeric	33	Number of satellites used
8	HDOP	-	numeric	0.6	Horizontal dilution of precision
9	ORTH	m	numeric	27.659	Orthometric height
11	GEOID	m	numeric	-	Geoid separation
13	AGE	s	numeric	-	Age of differential corrections
14	REFS_ID	-	numeric	-	ID of station providing differential corrections
15	NAV_IND	-	string	-	Navigation status indicator, refer to IEC-61108
16	cs	-	hexadecimal	*39	Checksum
17	<CR><LF>	-	character	-	Carriage return and line feed

© 2022-2024 Beken Corporation

2.2.10 NMEA_ZDA

Table 2-23 NMEA_ZDA

Message	NMEA_ZDA
NMEA Message ID	ZDA
Description	Time and date provided by the GNSS receiver
Firmware	-
Comment	-
Information	Number of fields： 8
Structure	$XXZDA,UTC,DAY,MONTH,YEAR,LTZH,LTZN*cs<CR><LF>
Example	$GNZDA,062632.00,04,02,24,00,00*7B<CR><LF>

Table 2-24 Payload of NMEA_ZDA

Field No.	Name	Unit	Format	Example	Description
0	$XXZDA	-	string	$GNZDA	Talker ID: GN for GNSS Message ID: ZDA
1	UTC	-	HHMMSS.ss	062632.00	UTC time of this epoch
2	DAY	day	dd	04	UTC date: day(range:1-31)
3	MONTH	month	mm	02	UTC date: month(range:1-12)
4	YEAR	year	yy	24	UTC date: year
5	LTZH	-	xx	00	Local time zone hours, always 00
6	LTZN	-	zz	00	Local time zone minutes, always 00
7	cs	-	hexadecimal	*7B	Checksum
8	<CR><LF>	-	character	-	Carriage return and line feed

2.2.11 NMEA_DTM

Table 2-25 NMEA_DTM

Message	NMEA_DTM

NMEA Message ID	DTM
Description	Outputs the offset between the Local and the Reference.
Firmware	X166X24385 and later versions
Comment	-
Information	Number of fields： 8
Structure	$XXDTM,LOC_CODE,SUB_CODE,LAT_O,DIR_LAT,LON_O,DIR_LON,ALT_O,REF_CODE*cs<CR><LF>
Example	$GNDTM,W84,,0.0,N,0.0,E,0.0,W84*71<CR><LF>

Table 2-26 Payload of NMEA_DTM

Field No.	Name	Unit	Format	Example	Description
0	$XXDTM	-	String	$GNDTM	Talker ID: GN for GNSS Message ID: DTM
1	LOC_CODE	-	XXX	W84	Local datum code: WGS84=W84 WGS72=W72 SGS85=S85 PE90=P90 user-defined
2	SUB_CODE	-	String		A null field
3	LAT_O	min	m.mm	0.0	Offset in Latitude
4	DIR_LAT	-	character	N	Direction of latitude: • N: North • S: South
5	LON_O	min	m.mm	0.0	Offset in Longitude
6	DIR_LON	-	character	E	Direction of longitude: • E: East • W: West
7	ALT_O	m	xx	0.0	Offset in altitude
8	REF_CODE	-	XXX	W84	Reference datum code: WGS84=W84 WGS72=W72

Field No.	Name	Unit	Format	Example	Description
					SGS85=S85 PE90=P90
9	cs	-	hexadecimal	*71	Checksum
10	<CR><LF>	-	character	-	Carriage return and line feed

2.2.12 NMEA_GBS

Table 2-27 NMEA_GBS

Message	NMEA_GBS
NMEA Message ID	GBS
Description	Output the results of the Autonomous Integrity Detection Algorithm (RAIM).
Firmware	X166X24385 and later versions
Comment	-
Information	Number of fields： 10
Structure	$XXGBS,UTC,ErrLat,ErrLon,ErrAlt,SVID,PROB,BIAS,STDdev,SYS_ID,SIG_ID *cs<CR><LF>
Example	$GNGBS,073252.00,2.06,6.03,7.64,12,,75.7,5.0,1,1,*75<CR><LF>

Table 2-28 Payload of NMEA_GBS

Field No.	Name	Unit	Format	Example	Description
0	$XXGBS	-	String	$GNGBS	Talker ID: GN for GNSS Message ID: GBS
1	UTC	-	HHMMSS.ss	073252.00	UTC time to which this RAIM sentence belongs
2	ErrLat	m	numeric	2.06	Expected error in latitude
3	ErrLon	m	numeric	6.03	Expected error in longitude
4	ErrAlt	m	numeric	7.64	Expected error in altitude
5	SVID	-	numeric	12	Satellite ID of most likely failed satellite
6	PROB	-	numeric	-	Probability of missed detection: null (not supported,fixed field)

© 2022-2024 Beken Corporation

Field No.	Name	Unit	Format	Example	Description
7	BIAS	m	numeric	75.7	Estimated bias of most likely failed satellite (a priori residual)
8	STDdev	m	numeric	5.0	Standard deviation of estimated bias
9	SYS_ID	-	hexadecimal	1	System ID, see Table 2-7.
10	SIG_ID	-	hexadecimal	1	NMEA-defined GNSS signal ID, See Table 2-7 (only available in NMEA 4.10 and later)
11	cs	-	hexadecimal	*75	Checksum
12	<CR><LF>	-	character	-	Carriage return and line feed

2.2.13 NMEA_GNS

Table 2-29 NMEA_GNS

Message	NMEA_GNS
NMEA Message ID	GNS
Description	Outputs Time, Position, and Information about GNSS positioning.
Firmware	X166X24385 and later versions
Comment	-
Information	Number of fields：13
Structure	$XXGNS,UTC,LAT,DIR_LAT,LON,DIR_LONG,POS_MODE,NUM_SV,HDOP,ALT,SEP,AGE,REFS_ID, NAV_ST*cs<CR><LF>
Example	$GNGNS,072414.00,3112.1673271,N,12135.9125501,E,AAAANA,22,1.2,16.591,0.000,,,V*29<CR><LF>

Table 2-30 Payload of NMEA_GNS

Field No.	Name	Unit	Format	Example	Description
0	$XXGNS	-	String	$GNGBS	Talker ID: GN for GNSS Message ID: GNS
1	UTC	-	HHMMSS.ss	072414.00	UTC time
2	LAT	-	DDMM.mmmmm	3112.1673271	Latitude
3	DIR_LAT	-	character	N	Direction of latitude:

Field No.	Name	Unit	Format	Example	Description
					• N: North • S: South
4	LON	-	DDDMM.mmmmm	12135.9125501	Longitude
5	DIR_LONG	-	character	E	Direction of longitude: • E: East • W: West
6	POS_MODE	-	numeric	AAAANA	The order of the six letters is： GPS、GLO、GAL、BDS、IRS、QZSS; For the meanings of letters, see Table Table 2-14 Position Fix Flags POS_MODE defined in the Locating standard.
7	NUM_SV	-	numeric	22	Number of satellites used
8	HDOP	-	numeric	1.2	Horizontal Dilution of Precision
9	ALT	m	numeric	16.591	Ellipsoidal Height
10	SEP	m	numeric	0.000	-
11	AGE	S	numeric	-	Age of differential corrections
12	REFS_ID	m	numeric	-	ID of station providing differential corrections
13	NAV_ST	-	character	V	S=Safe C=Caution U=Unsafe V=Navigational status not valid, equipment is not providing navigational status indication
14	cs	-	hexadecimal	*29	Checksum
15	<CR><LF>	-	character	--	Carriage return and line feed

2.3 PO Information Messages

Table 2-31 shows the proprietary information messages for the BK166X receiver. The PO messages are used for debugging purposes and can be disabled via user configuration.

© 2022-2024 Beken Corporation

Table 2-31 PO Information Messages

Message	Message ID	Description
Antenna Status	PO_ANT	Antenna status, open circuit or short circuit
Clock Source	PO_CLK	Clock information
Signal Strength	PO_CNR	Average C/N0 of the four strongest satellites for each constellation. The value is accurate to one decimal place.
Jamming	PO_JAM	Jamming information
Version	PO_LRS	Version information
Dead Reckoning Solution	PO_INS	Dead reckoning (DR) solution status and information
Odometer information	PO_ODO	Output current odometer information (cumulative displacement)
AGNSS Aiding data	PO_AID	Output the satellite ephemeris collected during the AGNSS process.
Received RTCM acknowledgement	PO_MSM	Output the age and number of observations of the difference data received in RTK mode.
Dual Antenna Data	PO_DUA	Output information transmitted from the board in dual antenna mode and inertial navigation inclination value.
Chip Info	BKCHIP	Output Chip information
IRNSS Message	PO_IRSF	Output IRNSS message information.
GAGAN Message	PO_GAMS	The GAGAN MSG63 message is displayed.
POTIM Message	PO_TIM	GPIO2 of BK166X outputs POTIM information after receiving the rising edge trigger.

2.3.1 Antenna Status

Table 2-32 NMEA_PO_ANT

Message	NMEA_PO_ANT
NMEA Message ID	PO_ANT
Description	Antenna status, open circuit, or short circuit.
Firmware	-
Comment	-
Information	Number of fields: 5
Structure	$POANT,antstat,antvol*cs<CR><LF>

© 2022-2024 Beken Corporation

Example	$POANT,1,803*4E

Table 2-33 Payload of NMEA_PO_ANT

Field No.	Name	Unit	Format	Example	Description
0	$POANT	-	string	$POANT	Message ID, BK protocol header, ANT sentence
1	antstat		numeric	1	• 0: Open circuit • 1: Normal • 2: Short circuit. Disable the output of EXT_LNA_LDO, and enable it 20 seconds later to detect the status again.
2	antvol	mV	numeric	803	Reserved
3	cs	-	hexadecimal	*5A	Checksum
4	<CR><LF>	-	character	-	Carriage return and line feed

2.3.2 Clock Source

Table 2-34 NMEA_PO_CLK

Message	NMEA_PO_CLK
NMEA Message ID	PO_CLK
Description	Clock information
Firmware	-
Comment	-
Information	Number of fields: 9
Structure	$POCLK,drift0,drift1,drift2,drift3,temp,data0*cs<CR><LF>
Example	$POCLK,2.414,2.410,-2.404,0.000,18.915,0*5A

Table 2-35 Payload of NMEA_PO_CLK

Field No.	Name	Unit	Format	Example	Description
0	$POCLK	-	string	$POCLK	Message ID, BK protocol header, CLK sentence
1	drift0	ppm	numeric	2.414	Clock drift per second, accurate to ppb
2	drift1	ppm	numeric	2.410	Internal data
3	drift2	ppm	numeric	-2.404	Internal data
4	drift3	ppm	numeric	0.000	Internal data
5	temp	°C	numeric	18.915	Chip internal temperature in degrees Celsius
6	data0	-	numeric	0	Internal data
7	cs	-	hexadecimal	*5A	Checksum
8	<CR><LF>	-	character	-	Carriage return and line feed

2.3.3 Signal Strength

Table 2-36 NMEA_PO_CNR

Message	NMEA_PO_CNR
NMEA Message ID	PO_CNR
Description	Average C/N0 of the four strongest satellites for each constellation. The value is accurate to one decimal place.
Firmware	-
Comment	-
Information	Number of fields: 13
Structure	$POCNR,data1,data2…data10*cs<CR><LF>
Example	$POCNR,3911,0,2945,0,0,3672,3519,3567,0,0*79

Table 2-37 Payload of NMEA_PO_CNR

Field No.	Name	Unit	Format	Example	Description
0	$POCNR	-	string	$POCNR	Message ID, BK protocol header, CNR statement
1-10	dataN	0.01 dB	numeric	3911	Average C/N0 of the four strongest satellites for each constellation. The value is accurate to one decimal place. • data0: GPS/QZSS/SBAS_L1 • data1: GPS/QZSS/SBAS_L5 • data2: BDS_B1I • data3: BDS_B2A • data4: BDS_B2B • data5: BDS_B1C • data6: GLO_G1 • data7: GAL_E1 • data8: GAL_E5A • data9: GAL_E5B
11	cs	-	hexadecimal	*77	Checksum
12	<CR><LF>	-	character	-	Carriage return and line feed

2.3.4 Jamming

Table 2-38 NMEA_PO_JAM

Message	NMEA_PO_JAM
NMEA Message ID	PO_JAM
Description	Receiver Jamming information
Firmware	-
Comment	-
Information	Number of fields: 12
Structure	$POJAM,L1rssi,L1gain,L5rssi,L5gain,L1spur_power,L1spur_freq,L5spur_power,L5spur_freq*cs <CR><LF>
Example	$POJAM,33,1844,35,1907,11,1600005,12,1186921*5C

© 2022-2024 Beken Corporation

Table 2-39 Payload of NMEA_PO_JAM

Field No.	Name	Unit	Format	Example	Description
0	$POJAM	-	string	$POJAM	Message ID, BK protocol header, JAM sentence
1	L1rssi	dB	numeric	33	Internal data If you need to use the data, please contact customer support.
2	L1gain	-	numeric	1844	Internal data If you need to use the data, please contact customer support.
3	L5rssi	dB	numeric	35	Internal data If you need to use the data, please contact customer support.
4	L5gain	-	numeric	1907	Internal data If you need to use the data, please contact customer support.
1	L1spur_power	-	numeric	11	Internal data If you need to use the data, please contact customer support.
2	L1spur_freq	kHz	numeric	1600005	Internal data If you need to use the data, please contact customer support.
3	L5spur_power	-	numeric	12	Internal data If you need to use the data, please contact customer support.
4	L5spur_freq	kHz	numeric	1186921	Internal data If you need to use the data, please contact customer support.
7	cs	-	hexadecimal	*5C	Checksum
8	<CR><LF>	-	character	-	Carriage return and line feed

2.3.5 Version

Table 2-40 NMEA_PO_LRS

Message	NMEA_PO_LRS

NMEA Message ID	PO_LRS
Description	Version information This information begins with `$POLARIS` in X166X23125 and earlier versions; The last five digits of the version can be used to distinguish the front and back of the version. The larger the number, the newer the version.
Firmware	-
Comment	Example: 4166124305RD The version number is defined as follows: • The 1st digit is an internal code; • The 2nd to 5th digits indicate the chip family supported by this version, such as 1616, 1661, and 1662; • The 6th and 7th digits indicate the year the version was released. As shown in the example, 24 represents 2024; • The 8th and 9th digits indicate the week (of the release year) the version was released. As shown in the example, 30 represents the 30th week of 2024. From 24205 version, add chars after version numbers which defined as below: • The char ‘R’, ‘I’ ,‘A’ or ’IR’ mean RTK, INS, Dual-Ant or RTK+INS version, and this char may be NULL; • The lasted char ‘S’ or ‘D’ mean Single freq or Dula freq.
Information	Number of fields: 5
Structure	$POLRS,version,data0*cs<CR><LF>
Example	$POLRS,4166124305RD,1*71

Table 2-41 Payload of NMEA_PO_RLS

Field No.	Name	Unit	Format	Example	Description
0	$POLRS	-	string	$POLRS	Message ID, BK protocol header, RLS sentence
1	version	-	string	166023125	Version number
2	data0	-	numeric	1	Internal data
3	cs	-	hexadecimal	*5A	Checksum
4	<CR><LF>	-	character	-	Carriage return and line feed

© 2022-2024 Beken Corporation

2.3.6 Dead Reckoning Solution

Table 2-42 NMEA_PO_INS

Message	NMEA_PO_INS
NMEA Message ID	PO_INS
Description	Dead reckoning (DR) solution status and information
Firmware	Version X166X2317X or later, Numbers 10-16 are valid for X166X24155 or later versions, Numbers 17-21 are valid for X166X24212 or later versions, Numbers 22-24 are valid for X166X24285 or later versions Numbers 25-27 are valid for X166X24365 or later versions
Comment	-
Information	Number of fields: 30
Structure	$POINS,GPS_week,GPS_seconds,INS_status,IMU_status,GNSS_status,odometer_status,motion_status,IMU_type,work_mode,roll,pitch,yaw,speed_status,lane_status,lean_status,bump_status,velicity_forward,velicity_rightward,velicity_downward,drive_mileage,work_time,acceler_forward,acceler_rightward,acceler_downward,angular_roll,angular_pitch,angular_yaw *cs<CR><LF>
Example	$POINS,2259,197506.000,2,1,1,0,1,3,1,0.036,0.141,0.382,0,0,1,0,14.75,0.02,0.04,20877.50,1757.46,1.5,0.2,0.5,-0.98,1.52,0.18*75

Table 2-43 Payload of NMEA_PO_INS

Field No.	Name	Unit	Format	Example	Description
0	$POINS	-	string	$POINS	Message ID, BK protocol header, INS sentence
1	GPS_week	-	numeric	-	GPS week number
2	GPS_seconds	sec	numeric	-	Seconds of GPS week
3	INS_status	-	digit	-	DR solution status: • 0: Not activated • 1: Initialization • 2: Installation status detection • 3: Initial alignment • 4: DR valid without convergence • 5: DR valid with convergence

© 2022-2024 Beken Corporation

Field No.	Name	Unit	Format	Example	Description
4	IMU_status	-	digit	-	IMU status: • 0: Data is invalid • 1: Data is normal • 2: Device data failure: data loss • 3: Device installation failure: no fixed carrier • 4: Device bias jump • 5: Device time error
5	GNSS_status	-	digit	-	GNSS status: • 0: Data is invalid • 1: Data is normal
6	odometer_status	-	digit	-	Odometer/wheel speed: • 0: Data is invalid • 1: Data is normal
7	motion_status	-	digit	-	Carrier movement status: • 0: Unknown • 1: Static • 2: Moving • 3: In linear motion • 4: In curvilinear motion
8	IMU_type	-	digit	-	IMU type: • 0: Unknown • 1: BMI270 • 2: ICM42680 • 3: LSM6DSR • 4: ICM42688P • 5: SCHA634-D03 • 6: LIS2DW12 • 7: ICM42607P • 8: ICM42670P • 9: BMI325
9	work_mode	-	digit	-	Operating mode: • 0: Invalid • 1: Vehicle mode • 2: Static test mode

Field No.	Name	Unit	Format	Example	Description
					• 3: Bicycle mode
10	roll	degrees	numeric	-	Carrier roll angle(range: -180–180)
11	pitch	degrees	numeric	-	Carrier pitch angle(range: -90–90)
12	yaw	degrees	numeric	-	Carrier yaw angle(range: -180–180)
13	speed_status	-	digit	-	Carrier speed status(Only valid in car mode): • 0: Normal • 1: Rapid acceleration • 2: Rapid deceleration
14	lane_status	-	digit	-	Carrier lane status(Only valid in car mode): • 0: Normal • 1: Sudden lane change • 2: Sharp turn
15	lean_status	-	digit	-	Carrier lean status(Only valid in car mode): • 0: Normal • 1: Lean
16	bump_status	-	digit	-	Carrier bumpiness status(Only valid in car mode): • 0: Normal • 1: Bump
17	velocity_forward	m/s	numeric	-	Carrier forward velocity
18	velocity_rightward	m/s	numeric	-	Carrier rightward velocity
19	velocity_downward	m/s	numeric	-	Carrier downward velocity
20	drive_mileage	meters	numeric	-	DR solution drive mileage
21	work_time	seconds	numeric	-	DR work time length in current INS status
22	acceler_forward	m/s^2	numeric	-	Carrier forward acceleration
23	acceler_rightward	m/s^2	numeric	-	Carrier rightward acceleration
24	acceler_downward	m/s^2	numeric	-	Carrier downward acceleration
25	angular_roll	degree/s	numeric	-	Roll angular velocity, as angular velocity of the carrier rotates around the forward axis, and positive and

© 2022-2024 Beken Corporation

Field No.	Name	Unit	Format	Example	Description
					negative values satisfy the right-hand screw rule
26	angular_pitch	degree/s	numeric	-	Pitch angular velocity, as angular velocity of the carrier rotates around the rightward axis, and positive and negative values satisfy the right-hand screw rule
27	angular_yaw	degree/s	numeric	-	Yaw angular velocity, as angular velocity of the carrier rotates around the downward axis, and positive and negative values satisfy the right-hand screw rule
28	cs	-	hexadecimal	*5A	Checksum
29	<CR><LF>	-	character	-	Carriage return and line feed

2.3.7 Odometer information

Table 2-44 NMEA_PO_ODO

Message	NMEA_PO_ODO
NMEA Message ID	Output odometer information (accumulative displacement)
Description	-
Firmware	-
Information	Number of fields: 4
Structure	$POODO,ODOMETER*cs<CR><LF>
Example	$POODO,0.001*15

Table 2-45 Payload of NMEA_PO_ODO

Field No.	Name	Unit	Format	Example	Description
0	$POODO,	-	string	$POODO,	Message ID, BK protocol header, ODO sentence
1	ODOMETER	Km	numeric	0.001	Output odometer information (accumulative displacement)
2	cs		hexadecimal	*15	Checksum

© 2022-2024 Beken Corporation

Field No.	Name	Unit	Format	Example	Description
3	<CR><LF>	-	character	--	Carriage return and line feed

2.3.8 AGNSS Aiding data

Table 2-46 NMEA_PO_AID

Message	NMEA_PO_AID
NMEA Message ID	Output a bitmask showing the satellite IDs of the received satellite ephemeris.
Description	Only active in AGNSS process。
Firmware	-
Information	Number of fields: 8
Structure	$POAID,bm0,bm1,bm2,bm3,bm4*cs<CR><LF>
Example	$POAID,0,0,0,0,0*4F

Table 2-47 Payload of NMEA_PO_AID

Field No.	Name	Unit	Format	Example	Description
0	$POAID,	-	string	$POAID,	Message ID, BK protocol header, AID sentence
1	bm0	-	numeric	1	Bitmask of GPS (decimal)
2	bm1	-	numeric	2	Bitmask of BDS3 (decimal)
3	bm2	-	numeric	4	Bitmask of BDS2 (decimal)
4	bm3	-	numeric	8	Bitmask of GLONASS (decimal)
5	bm4	-	numeric	16	Bitmask of Galileo (decimal)
6	cs	-	hexadecimal	*15	Checksum
7	<CR><LF>	-	character	--	Carriage return and line feed

2.3.9 Received RTCM acknowledgement

Table 2-48 NMEA_PO_MSM

Message	NMEA_PO_MSM

NMEA Message ID	Output received RTCM MSM messages information during RTK mode.
Description	Support RTCM MSM4 and MSM7
Firmware	-
Information	Number of fields:20
Structure	$POMSM,t0,nsig0,nsat0,t1,nsig1,nsat1,t2,nsig2,nsat2,t3,nsig3,nsat3,t4,nsig4,nsat4,d0,d1*cs<CR><LF>
Example	$POMSM,167,2,05,167,2,14,000,0,00,167,2,03,000,0,00,9,1*42

Table 2-49 Payload of NMEA_PO_MSM

Field No.	Name	Unit	Format	Example	Description
0	$POMSM,	-	string	$POMSM,	Message ID
1	t0	sec	numeric	167	Receiving time mark for GPS MSM4(Low 8bits of week seconds).
2	nsig0	-	numeric	2	Signals number of receiving GPS MSM4.
3	nsat0	-	numeric	05	Satellites number of receiving GPS MSM4.
4	t1	sec	numeric	167	Receiving time mark for BDS3 MSM4(Low 8bits of week seconds).
5	nsig1	-	numeric	2	Signals number of receiving BDS3 MSM4.
6	nsat1	-	numeric	05	Satellites number of receiving BDS3 MSM4.
7	t2	sec	numeric	167	Receiving time mark for GLONASS MSM4(Low 8bits of week seconds).
8	nsig2	-	numeric	2	Signals number of receiving GLONASS MSM4.
9	nsat2	-	numeric	05	Satellites number of receiving GLONASS MSM4.
10	t3	sec	numeric	167	Receiving time mark for Galileo MSM4(Low 8bits of week seconds).
11	nsig3	-	numeric	2	Signals number of receiving Galileo MSM4.
12	nsat3	-	numeric	05	Satellites number of receiving Galileo MSM4.

Field No.	Name	Unit	Format	Example	Description
13	t4	sec	numeric	167	Receiving time mark for BDS2 MSM4(Low 8bits of week seconds).
14	nsig4	-	numeric	2	Signals number of receiving BDS2 MSM4.
15	nsat4	-	numeric	05	Satellites number of receiving BDS2 MSM4.
16	d0	-	-	-	Internal data
17	d1	-	-	-	Internal data
18	cs	-	hexadecimal	*15	Checksum
19	<CR><LF>	-	character	--	Carriage return and line feed

2.3.10 Dual Antenna Data

Table 2-50 NMEA_PO_DUA

Message	NMEA_PO_DUA
NMEA Message ID	PODUA
Description	Output data received from slave chip and sensor tilt.
Firmware	DUALANT mode
Information	Number of fields: 14
Structure	$PODUA,d1,d2,d3,d4,d5 nsat1,nsat2,nsat3,nsat4,nsat5,d6,d7,d8,tilt*cs<CR><LF>
Example	$POMSM,0,0,0,0,0,10,4,8,3,6,255,4.5,8,-3.9,1*42

Table 2-51 Payload of NMEA_PO_DUA

Field No.	Name	Unit	Format	Example	Description
0	$PODUA,	-	string	$PODUA,	Message ID
1	d1	-	-	-	Internal data
2	d2	-	-	-	Internal data
3	d3	-	-	-	Internal data
4	d4	-	-	-	Internal data

Field No.	Name	Unit	Format	Example	Description
5	d5	-	-	-	Internal data
6	nsat1	-	numeric	10	GPS satellite number received from slave chip
7	nsat2	-	numeric	4	BDS3 satellite number received from slave chip
8	nsat3	-	numeric	8	GLONASS satellite number received from slave chip
9	nsat4	-	numeric	3	Galileo satellite number received from slave chip
10	nsat5	-	numeric	6	BDS2 satellite number received from slave chip
11	d6	-	-	-	Internal data
12	d7	-	-	-	Internal data
13	d8	-	-	-	Internal data
14	tilt	-	numeric	-3.9	Sensor tilt
15	cs	-	hexadecimal	*42	Checksum
16	<CR><LF>	-	character	--	Carriage return and line feed

2.3.11 Chip Info

Table 2-52 NMEA_BKCHIP

Message	NMEA_BKCHIP
NMEA Message ID	BKCHIP
Description	Chip information
Firmware	-
Comment	Chip version include 1661G,1661S,1662,1662G,1662T,1662G_C,1662G_I,1662N,1662N_S,1662N_D
Structure	$BKCHIP,chipinfo*cs
Example	$BKCHIP,1662G*73

2.3.12 IRNSS Message

Table 2-53 NMEA_PO_IRNSF

Message	NMEA_PO_IRSF
NMEA Message ID	POIRSF
Description	Output IRNSS message information.
Firmware	4166024275 and later versions
Information	Number of fields: 39
Structure	$POIRSF,sv,sf,d0,d1,d2,…,d35,d36*cs<CR><LF>
Example	$POIRSF,3,2,8B,0F,B5,49,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,55,56,3D,B1,1C,0*0F

Table 2-54 Payload of NMEA_PO_IRNSF

Field No.	Name	Unit	Format	Example	Description
0	$POIRNSF,	-	string	$POIRNSF,	Message ID
1	sv	-	numeric	3	Satellite number
2	sf	-	numeric	2	Subframe number
3-38	d0-d35	-	hexadecimal	8B,…,1C	Subframe data, starting from TLM
39	d36	-	hexadecimal	0	Subframe end data
40	cs	-	hexadecimal	0F	Checksum
41	<CR><LF>	-	character	--	Carriage return and line feed

2.3.13 GAGAN Message

Table 2-55 NMEA_PO_GAMS

Message	NMEA_PO_GAMS
NMEA Message ID	POGAMS
Description	Output GAGAN MSG63 message information.
Firmware	4166024275 and later versions

Information	Number of fields: 28
Structure	$POGAMS,sv,d0,d1,d2,…,d25,d26*cs<CR><LF>
Example	$POGAMS,45,00,40,00,04,2C,99,FA,40,00,08,30,05,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00,00*7A

Table 2-56 Payload of NMEA_PO_GAMS

Field No.	Name	Unit	Format	Example	Description
0	$POGAMS,	-	string	$POGAMS,	Message ID
1	sv	-	numeric	45	Satellite number
2-28	d0-d25	-	hexadecimal	00,⋯,00	Type63 message bit flow
29	d26	-	hexadecimal	00	Type63 message bit flow， Low 4 bit zeros
30	cs	-	hexadecimal	7A	Checksum
31	<CR><LF>	-	character	--	Carriage return and line feed

2.3.14 POTIM Message

Table 2-57 NMEA_PO_TIM

Message	NMEA_PO_TIM
NMEA Message ID	POTIM
Description	The BK166X outputs POTIM information after GPIO2 receiving the rising edge trigger.
Firmware	4166024125 and later versions.
Information	Number of fields: 7
Structure	$ POTIM,Flag,Timebase,Count,Wn,TowMs,TowSubMs,AccEst*cs<CR><LF>
Example	$POTIM,3,0,185,2326,282893000,15,20*65

Table 2-58 Payload of NMEA_PO_TIM

Field No.	Name	Unit	Format	Example	Description
0	$POTIM,	-	string	$ POTIM,	Message ID
1	Flag	-	numeric	3	Bit0:

Field No.	Name	Unit	Format	Example	Description
					0=Time is not valid 1=Time is valid (Valid GNSS fix)
2	Timebase	-	numeric	0	0:GPS Time 1:BDS Time 2:GLO Time 3:GAL Time GPS Time by default
3	Count	-	numeric	185	Rising edge counter
4	Wn	week	numeric	2326	Week number of last rising edge
5	TowMs	ms	numeric	282893000	Tow of last rising edge
6	TowSubMs	ns	numeric	15	Millisecond fraction of tow of rising edge in nanoseconds
7	AccEst	ns	numeric	20	Accuracy estimate
8	cs	-	hexadecimal	65	Checksum
9	<CR><LF>	-	character	--	Carriage return and line feed

2.4 PO_NMEA Commands

2.4.1 PO_NMEA Command Format

The basic format of the PO_NMEA command is: `$POLMSGNMEA,data1,data2,data3…\r\n`(X166X24115 and later no need to add '\r\n' at the end of the command)

Example: `$POLCFGMSG,1,1\r\n`

All PO_NMEA commands start with '`$POLCFG`' (0x24504F434647) followed by the command name, then followed by an indefinite number of parameters or data. The command name and data are separated by commas (0x2C). Letters in command names and parameters are all uppercase.

When the receiver receives the command correctly, it will return '`$OK`', otherwise it will return '`$FAIL`'.

Note: When using the serial port tool to send commands, a carriage return and line feed should be added at the end. When using the NavStarTool tool, it automatically adds '`\r\n`' (0x0D0A) at the end of the command.

© 2022-2024 Beken Corporation

2.4.2 Restart

Message Format	$POLCFGRESET,Type	
Example	$POLCFGRESET,1	
Description	Receiver reset, It becomes effective immediately	
Message Type	Input	
Parameter	Type	Description
Type	uint8	Reset type • 0: Hot start • 1: Cold start

2.4.3 Baud Rate Configuration

Message Format	$POLCFGPRT,baud,reserved	
Example	$POLCFGPRT,115200,0	
Description	This command is used to configure the port baud rate. It becomes effective immediately. This configuration could be saved by 2.4.9 save to flash command.	
Message Type	Input	
Parameter	Type	Description
baud	uint8	Port baud rate, options: 9600, 38400, 57600, 115200, 230400, 460800, 1000000, 2000000
reserved	uint8	Reserved

Note: Since the corresponding baud rate of the host has not been changed, '`$OK`' or '`$FAIL`' will not be received.

2.4.4 Configuration of Solution Update Rate and Observation Update Rate

Message Format	$POLCFGNAV,navRate,measRate
Example	$POLCFGNAV,10,5

© 2022-2024 Beken Corporation

Description	This command is used to configure PVT (Position, Velocity and Time) solution update rate and observation update rate. It will become effective after 2 seconds. This configuration could be saved by 2.4.9 save to flash command.	
Message Type	Input/output	
Parameter	Type	Description
navRate	uint8	PVT solution update rate, options: 1, 5, 10, 20 (unit: Hz)
measRate	uint8	Observation update rate, options: 1, 5, 10, 20 (unit: Hz)

2.4.5 NMEA Message Output Configuration

Message Format	$POLCFGMSG,msgClass(0),msgID,msgCtrl $POLCFGMSG,msgClass(1),msgRate $POLCFGMSG,msgClass(2),msgID,msgRate,msgOrder
Example	$POLCFGMSG,0,2,0 $POLCFGMSG,1,5 $POLCFGMSG,2,5,5,1
Description	X166X24451 and later versions support the related configuration with the first field is 2. This command is used to Configure NMEA output switch, rate, order: When the first field is 0, then enter msgID, msgCtrl and the output switch that only controls a single NMEA cannot be used to control the output frequency. It becomes effective immediately, this configuration could be saved by 2.4.9 save to flash command. In this case, msgCtrl can only select two values, 0 or 1. Example: `$POLCFGMSG,0,2,0` disable GSV sentence output Example: `$POLCFGMSG,0,2,1` enable GSV sentence output When the first field is 1, then enter msgRate, which is used to control the switching or output frequency of all NMEA. It will become effective after 2 seconds. This configuration save to flash command automatically. Example: `$POLCFGMSG,1,5` change the NMEA output rate to 5 Hz Example: `$POLCFGMSG,1,0` disable the output of all NMEA sentences

	When the first field is 2, msgID, msgRate and msgOrder are entered after it, indicating that the output rate and output order of the NMEA statement corresponding to msgID are adjusted. Note that the current msgID support GGA/GSA/GSV/vacations/RMC/GST/GLL GFA/ZDA/GMP. Example: $POLCFGMSG,2,5,5,1 Change the output rate of the RMC statement to 5Hz, and the output order is adjusted to the first.	
Message Type	Input/output	
Parameter	Type	Description
msgClass	uint8	Optional, 0,1,2 • 0: Control the output of a certain NMEA sentence type • 1: Control the output rate of all NMEA output sentences or disable the output of all NMEA sentences • 2: Control the output rate and order of a certain NMEA sentence type
msgID	uint16	• 0: GGA • 1: GSA • 2: GSV • 3: VTG • 4: CNR • 5: RMC • 6: CLK • 7: POL • 8: THS • 9: ANT • 10: JAM • 11: INS • 12: GST • 13: GLL • 14: AID • 15: MSM

		• 16: GMP
msgCtrl/msgRate	uint8	When msgClass=0, it means enabling or disabling the output of a certain NMEA sentence individually. 0: disable; 1: enable When msgClass=1, if msgRate is configured to 1, 5, 10, or 20 (unit: Hz), it means controlling the output rate of current NMEA output sentences; if msgRate is configured to 0, the output of all NMEA sentences will be disabled.

2.4.6 Constellation Signal Configuration

Message Format	$POLCFGSYS,sysMask	
Example	$POLCFGSYS,11	
Description	This command is used to configure the constellation signals. It will become effective after 2 seconds. This configuration could be saved by 2.4.9 save to flash command.	
Message Type	Input	
Parameter	Type	Description
sysMask	uint32	Systems signal mask If the corresponding bit is set to 1, the signal is enabled. • Bit 0: GPS_L1 • Bit 1: GPS_L5 • Bit 2: BDS_B1I • Bit 3: BDS_B2A • Bit 4: BDS_B2B • Bit 5: BDS_B1C • Bit 6: GLO_G1 • Bit 7: GAL_E1 • Bit 8: GAL_E5A • Bit 9: GAL_E5B • Bit 13: IRS_L5 • Bit 14: BD2_B2I

		1. When entered in decimal: $POLCFGSYS,39 Configure signals GPS_L1,GPS_L5,BDS_B1I, and BDS_B1C 2. When entered in hexadecimal, end with an 'H' $POLCFGSYS,27H Equal to $POLCFGSYS,39 Configure the signal as GPS_L1,GPS_L5,BDS_B1I,BDS_B1C 3. When ending with '+' or '-', it means that a signal is turned on or off based on the original signal. At this time, the input bit corresponding to the signal can be '+' or '-'. For example, $POLCFGSYS,5+ '5' corresponds to the BDS_B1C bit. This command enables the B1C signal on the basis of the original signal For example, $POLCFGSYS,6- '6' corresponds to the GLO_G1 bit. This command disables the G1 signal on the basis of the original signal.

2.4.7 RTCM Configuration

Message Format	$POLCFGRTCM,msmType,msmRate,ephFlag	
Example	$POLCFGRTCM,0,5,0	
Description	This command is used to configure RTCM. It becomes effective immediately. This configuration could be saved by 2.4.9 save to flash command. Note: The msmRate shall be equal or less than receiver working rate or fix rate.	
Message Type	Input/output	
Parameter	Type	Description
msmType	uint8	RTCM message type: • 0: MSM7 • 1: MSM4

msmRate	uint8	If it is set to 0, all RTCM output is disabled. The msmType and ephFlag will be omitted. RTCM output rate, options: 0, 1, 5, 10, 20 (unit: Hz. This rate shall be equal or less than receiver working rate or fix rate)
ephFlag	uint8	• 0: Disable ephemeris output • 1: Enable ephemeris output

2.4.8 Query Constellations Configuration

Message Format	$POLCFGPTSYS
Example	$POLCFGPTSYS
Description	This command queries the current constellation signal and returns active constellation bitmask. For example, `$POSYS_BM:0x000021EF`
Message Type	Input/output

2.4.9 Save Configuration

Message Format	$POLCFGSAVE
Example	$POLCFGSAVE
Description	This command is used to save current configuration. It becomes effective immediately.
Message Type	Input

2.4.10 Base station coordinates Configuration

Message Format	$POLCFGBASE,latitude,longitude, height
Example	$POLCFGBASE,30.064146,106.228056,21.15
Description	This command is used to configure the base station coordinates. It becomes effective immediately. This configuration could be saved by 2.4.9 save to flash command.

Message Type	Input	
Parameter	Type	Description
latitude	double	Base station latitude
longitude	double	Base station longitude
height	double	Base station height

2.4.11 PPS Configuration

Message Format	$POLCFGSTS,stsMode,pulesMode, outputType,pulesInterval,pulesWidth,cableDelay	
Example	$POLCFGSTS,1,0,0,1000,10000,-100	
Description	The PPS configuration takes effect after 2 seconds. This configuration could be saved by 2.4.9 save to flash command.	
Message Type	Input	
Parameter	Type	Description
stsMode	uint8	stsMode High-precision timing chip works. 0:Disable 1:Normal Static Mode 2:Set Position Mode(Fix Position) 3:Single Sat Mode(Fix Position)
pulesMode	uint8	0:Rising Edge 1:Falling Edge
outputType	uint8	0:Accurate Time 1:Always 2:Disabled
pulesInterval	uint32	PPS interval, configurable range (0,100000)
pulesWidth	uint32	PPS width, configurable range (0,100000000)
cableDelay	int16	PPS delay, configurable range (-32767,32767)

2.4.12 Get Receiver Version

Message Format	$POLCFGPTVER
Example	$POLCFGPTVER
Description	X166X24335 and later versions
Message Type	Input

2.4.13 Configure the Test Mode of Product Line

Message Format	$POLCFGATE,mode,param0	
Example	$POLCFGATE,1,11	
Description	The configuration takes effect immediately. Flash cannot be saved. The configuration becomes invalid after a reset.	
Message Type	Input	
Parameter	Type	Description
mode	uint8	0: ATE mode off 1: ATE CW output(After configuring this mode, the detected CW signal Power has been in the form of CNR output in GSV).
param0	uint8	In ATE CW output mode, it is defined as the detection of CW frequency dot mask: Bit0: L1 1575.42MHz Bit1: B1 1561.098MHz Bit2: G1 1602MHz Bit3: L5 1176.45MHz Bit4: B2 1207.14MHz Example: Detect L1： $POLCFGATE,1,1 Detect L1+B1+L5: $POLCFGATE,1,11 Detect all frequencies： $POLCFGATE,1, 31

2.4.14 Configuring the Mincnr Value

Message Format	$POLCFGMCNR,mincnr	
Example	$POLCFGMCNR,30	
Description	Set the minimum CNR value to participate in the solution. Satellites below mincnr will not participate in the positioning solution; It becomes effective immediately. This configuration could be saved by 2.4.9 save to flash command; X166X24385 and later versions	
Message Type	Input	
Parameter	Type	Description
mincnr	uint8	Optional range: 5-40dB

2.4.15 Configure GNSS Antenna Lever Arm of Integrated Navigation

Message Format	$POLCFGIGLA,lever_arm_x,lever_arm_y,lever_arm_z	
Example	$POLCFGIGLA,0.83,0.21,-0.52	
Description	Configure the GNSS antenna lever arm, send the save command in 2.4.9 and it will take effect only after power off and restart. This command is effective in versions X166X24445 and later.	
Message Type	Input	
Parameter	Type	Description
lever_arm_x	double	The component of the lever arm in the forward direction of the Carrier coordinate system, range [-10, 10], unit meter
lever_arm_y	double	The component of the lever arm in the rightward direction of the Carrier coordinate system, range [-10, 10], unit meter
lever_arm_z	double	The component of the lever arm in the downward direction of the Carrier coordinate system, range [-10, 10], unit meter

2.4.16 Configure Wheel Odometer Lever Arm of Integrated Navigation

Message Format	$POLCFGIOLA,lever_arm_x,lever_arm_y,lever_arm_z	
Example	$POLCFGIOLA,-1.81,0.24,0.90	
Description	Configure the Wheel Odometer lever arm, send the save command in 2.4.9 and it will take effect only after power off and restart. This command is effective in versions X166X24445 and later.	
Message Type	Input	
Parameter	Type	Description
lever_arm_x	double	The component of the lever arm in the forward direction of the Carrier coordinate system, range [-10, 10], unit meter
lever_arm_y	double	The component of the lever arm in the rightward direction of the Carrier coordinate system, range [-10, 10], unit meter
lever_arm_z	double	The component of the lever arm in the downward direction of the Carrier coordinate system, range [-10, 10], unit meter

2.4.17 Configure IMU Mounted Axis of Integrated Navigation

Message Format	$POLCFGICAX,axis_forward,axis_rightward, axis_downward	
Example	$POLCFGICAX,-1,-2,3	
Description	Configure the IMU mounted axis, send the save command in 2.4.9 and it will take effect only after power off and restart. This command is effective in versions X166X24445 and later.	
Message Type	Input	
Parameter	Type	Description
axis_forward	int8	The mounted relationship between the carrier forward axis and the IMU XYZ axis is as follows: 1: IMU X-axis;

© 2022-2024 Beken Corporation

		-1: IMU -X-axis; 2: IMU Y-axis; -2: IMU -Y-axis; 3: IMU Z-axis; -3: IMU -Z-axis
axis_rightward	int8	The mounted relationship between the carrier rightward axis and the IMU XYZ axis
axis_downward	int8	The mounted relationship between the carrier downward axis and the IMU XYZ axis

2.4.18 Configure IMU Mounted Misalignment of Integrated Navigation

Message Format	$POLCFGICMS,misalignment_roll,misalignment_pitch, misalignment_yaw	
Example	$POLCFGICMS,-3.2,2.56,7.31	
Description	Configure the IMU mounted misalignment, send the save command in 2.4.9 and it will take effect only after power off and restart. This command is effective in versions X166X24445 and later.	
Message Type	Input	
Parameter	Type	Description
misalignment_roll	double	Based on the carrier's front-right-down right-hand system, the misalignment angle of the IMU around the forward axis, range [-45, 45], unit degree
misalignment_pitch	double	Based on the carrier's front-right-down right-hand system, the misalignment angle of the IMU around the rightward axis, range [-45, 45], unit degree
misalignment_yaw	double	Based on the carrier's front-right-down right-hand system, the misalignment angle of the IMU around the downward axis, range [-45, 45], unit degree

2.4.19 Configure Debug Log Output of Integrated Navigation

Message Format	$POLCFGIDBG,debug_mode

Example	$POLCFGIDBG,1	
Description	Configure debug log output of Integrated Navigation, which takes effect immediately. After it takes effect, you can save it permanently using the 2.4.9 save command. This command is effective in versions X166X24445 and later.	
Message Type	Input	
Parameter	Type	Description
debug_mode	uint8	Debug log output switch: • 0: Disable • 1: Enable

3. BK Protocol

3.1 BK Protocol Key Features

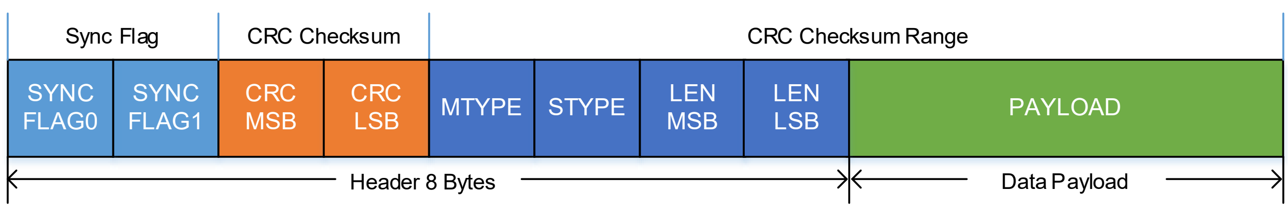
The BK166X receiver supports a Beken-proprietary protocol to communicate with a host. This protocol has the following key features:

- Compact – uses 8-bit binary data
- CRC protected – uses a CRC-16 algorithm
- Modular – uses a two-stage message identifier (MTYPE and STYPE)

3.2 BK Protocol Frame Structure

Figure 3-1 shows the structure of a BK protocol message.

Figure 3-1 BK Protocol Frame Structure



Every Frame starts with an 8-byte header and the payload follows. The number format of the frame structure is little-endian.

Table 3-1 BK Protocol Frame Structure

Byte Order	Field	Description
0	SYNC FLAG0	Synchronization flag 0, always 0x42 (ASCII 'B')
1	SYNC FLAG1	Synchronization flag 1, always 0x4B (ASCII 'K')
2	CRC MSB	8 MSB bits of 2-byte CCITT CRC16 checksum. The CRC checksum range is illustrated in Figure 3-1.
3	CRC LSB	8 LSB bits of 2-byte CCITT CRC16 checksum
4	MTYPE	Main type of a 2-byte/2-stage message identifier field
5	STYPE	Sub-type of a 2-byte/2-stage message identifier field
6	LEN MSB	Upper 4 bits indicate the sequence in which message are sent. The sequence automatically increases by 1 after each message is sent. Lower 4 bits indicate 4 MSB bits of payload length.

© 2022-2024 Beken Corporation

Byte Order	Field	Description
7	LEN LSB	8 LSB bits of payload length. The payload length field does not include the SYNC FLAG, CRC, MTYPE, STYPE, or LEN fields.
8 to [LEN +7]	PAYLOAD	The payload field contains a variable number of bytes (range: 0–1024).

3.3 Acknowledgement (ACK) Message

When a message is sent to the receiver, the receiver will send an ACK (acknowledgement) message back to the host, indicating the message has been received correctly.

The ACK message has the same format as the received message, except that its MTYPE value will be increased by 1 based on the received message. Its STYPE value is the same as the received message, and its data segment length may be empty.

3.4 PNT Message

3.4.1 NAV Messages

Table 3-2 BK_PNT_NAV

Message	BK_PNT_NAV				
Description	Outputs include position, time, speed, satellite information.				
Firmware	Version X166X24335 or later				
Comment					
Message Structure	Header	MTYPE	STYPE	LEN (Bytes)	
	0x42 0x4B	0x41	0x0	60	
Payload description:					
Byte Offset	Number Format	Scale	Name	Unit	Description
0	uint16	-	millisecond	ms	Milliseconds of second, range 0..999 (UTC)
2	uint8	-	second	sec	Seconds of minute, range 0..59 (UTC)
3	uint8	-	minute	min	Minute of hour, range 0..59 (UTC)

4	uint8	-	hour	h	Hour of day, range 0..23 (UTC)
5	uint8	-	day	d	Day of month, range 1..31 (UTC)
6	uint8	-	month	mon	Month, range 1..12 (UTC)
7	uint8	-	year	y	Year(From 2000) (UTC), example 24 for 2024
8	uint8	-	flags	-	Bits 0…2 0 = no fix 1 = dead reckoning only 2 = 2D-fix 3 = 3D-fix 4 = GNSS + dead reckoning combined 5 = SABS or RTD fix 6 = RTK float 7 = RTK fix
					Bit 3 0 = sensor not installed or misfunction 1 = sensor function enabled
					Bits 4…5 0 = no differential input above 120 sec 1 = differential input less than 5 sec 2 = differential input between 5-120 sec 3 = not defined
					Bits 6…7 0 = Time not valid 1 = Date valid 2 = Coarse time valid 3 = Fine time valid

9	uint8	-	numSV	-	Number of satellites used in Nav Solution
10	uint8	0.1	pDOP	-	Position DOP
11	uint8	-	Version		Version of PNT
12	uint16	-	GNSS system bitmap	-	GNSS signal indication: If the corresponding bit is set to 1, the signal is enabled. And it will output the average C/N0 and svNumber of the signal. (Maximum output of 8 signals) Bit0:GPS_L1 Bit1:GPS_L5 Bit2:GPS_L2C Bit3:BDS_B1I Bit4:BDS_B2A Bit5:BDS_B2B Bit6:BDS_B3I Bit7:BDS_B1C Bit8:BDS_B2I Bit9:GLO_G1 Bit10:GLO_G2 Bit11:GAL_E1 Bit12:GAL_E5A Bit13:GAL_E5B Bit14:IRS_L5 Bit15:SBS_L1
14	uint8	-	C/N0	dB	If the signal is enable, it will output average C/N0. Average C/N0 of the four strongest satellites for each constellation.

22	uint8	-	svNumber	-	If the signal is enable according to GNSS signal indication, it will output used sv number of this signal.
30	float16	-	hAcc	m	Horizontal accuracy estimate
32	float16	-	vAcc	m	Vertical accuracy estimate
34	float16	-	sAcc	m/s	Speed accuracy estimate
36	float16	-	hAcc	deg	Heading accuracy estimate (both motion and vehicle)
38	float16	-	velN	m/s	NED north velocity
40	float16	-	velE	m/s	NED east velocity
42	float16	-	velD	m/s	NED down velocity
44	float16	-	gSpeed	m/s	Ground Speed (2-D)
46	float16	-	head	deg	Heading of motion (2-D)
48	int32	-	longitude	deg	Longitude
52	int32	-	latitude	deg	Latitude
56	int32	-	height	mm	Height above ellipsoid

NOTE:float16 means IEEE754-2008 half floating data format. The standard mandates binary floating point data be encoded on three fields: a one bit sign field, followed by 5 exponent bits encoding the exponent offset by a numeric bias specific to each format, and 10 bits encoding the significand (or fraction).

3.5 Receiver Configuration Commands

To configure the serial port baud rate, output message, IO port, debug mode, receiver application scenarios, output frequency and enabling/disabling of other functions, it is recommended to use the relevant commands provided in this section.

There are two configuration commands, one for setting and one for getting a specific receiver configuration parameter.

Table 3-3 Receiver Configuration Commands

Message	Message ID	Description
Set receiver configuration	BK_USRCFG_SET	Sets a specific receiver configuration parameter.
Get receiver configuration	BK_USRCFG_GET	Gets a specific receiver configuration parameter.

3.5.1 BK_USRCFG_SET

The BK_USRCFG_SET command is used to configure a specific receiver parameter.

Each receiver configuration should include the configuration item ID and its value.

Note: After configuring the receiver, the user should wait for the receiver to start operating properly before performing subsequent operations.

Table 3-4 BK_USRCFG_SET

<table>
<tr><td colspan="2">Message</td><td colspan="4">BK_USRCFG_SET</td></tr>
<tr><td colspan="2">Description</td><td colspan="4">This command is used to set a specific receiver configuration parameter.</td></tr>
<tr><td colspan="2">Firmware</td><td colspan="4">Version X166X22475 or later</td></tr>
<tr><td colspan="2">Comment</td><td colspan="4">This command provides a simple way to set a single receiver configuration parameter by providing a configuration item ID-Value pair. The receiver will reboot after successful setting.
Note: It is recommend to use the GenCmdHex executable file to generate this command.</td></tr>
<tr><td colspan="2" rowspan="2">Message Structure</td><td>Header</td><td>MTYPE</td><td>STYPE</td><td>LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x26</td><td>0x02</td><td>12</td></tr>
<tr><td colspan="6">Payload description:</td></tr>
<tr><td>Byte Offset</td><td>Number Format</td><td>Scale</td><td>Name</td><td>Unit</td><td>Description</td></tr>
<tr><td>0</td><td>uint32</td><td>-</td><td>Version Identifier</td><td>-</td><td>Reserved</td></tr>
<tr><td>4</td><td>uint32</td><td>-</td><td>Config ID</td><td>-</td><td>Configuration item ID
For the meaning of the configuration item ID, refer to the configuration template JASON file.</td></tr>
<tr><td>8</td><td>uint32</td><td>-</td><td>Parameter Value</td><td>-</td><td>Configuration item value
For the valid range and meaning of the parameter value, refer to the configuration template JASON file.</td></tr>
</table>

3.5.2 BK_USRCFG_GET

The BK_USRCFG_GET command is used to read a specific receiver configuration parameter. After receiving this command, the receiver returns its internal corresponding configuration to the host.

Getting a specific receiver configuration parameter requires the configuration item ID. The BK_USRCFG_GET command does not restart the receiver, so the host can perform subsequent operations as soon as it receives the response (ACK) to the command.

Table 3-5 BK_USRCFG_GET

<table>
<tr><td>Message</td><td colspan="6">BK_USRCFG_GET</td></tr>
<tr><td>Description</td><td colspan="6">This command is used to gets a specific receiver configuration parameter.</td></tr>
<tr><td>Firmware</td><td colspan="6">Version X166X22475 or later</td></tr>
<tr><td>Comment</td><td colspan="6">The command provides a simple way to get a single receiver configuration parameter by providing a configuration item ID. The receiver returns the parameter value in the ACK message.
Note: It is recommend to use the GenCmdHex executable file to generate this command.</td></tr>
<tr><td rowspan="2">Message Structure</td><td>Header</td><td>MTYPE</td><td>STYPE</td><td colspan="3">LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x26</td><td>0x03</td><td colspan="3">8</td></tr>
<tr><td colspan="7">Payload description:</td></tr>
<tr><td>Byte Offset</td><td>Number Format</td><td>Scale</td><td>Name</td><td>Unit</td><td colspan="2">Description</td></tr>
<tr><td>0</td><td>uint32</td><td>-</td><td>Version Identifier</td><td>-</td><td colspan="2">Reserved</td></tr>
<tr><td>4</td><td>uint32</td><td>-</td><td>Config ID</td><td>-</td><td colspan="2">Configuration item ID
For the meaning of the configuration item ID, refer to the configuration template JASON file.</td></tr>
</table>

3.6 General Messages

General messages are typically used for one-time commands or instruction requests, including Start, Sleep, and one-time baud rate modification, etc. Commonly used general messages are shown in Table 3-5. General message content is not saved to the Flash memory. However, some commands can clear or modify the content in the Flash memory of the receiver.

Table 3-6 Commonly Used General Messages

Message Name	Message ID	Description
Deep Sleep	BK_PWR_DeepSleep	Makes the receiver enter deep sleep mode to save power and set wakeup source.
Stop Receiver	BK_CTRL_Stop	Stops the operation of the receiver.
Start Receiver	BK_CTRL_Start	Starts the receiver (if it stops operating).

Message Name	Message ID	Description
Restart Receiver	BK_CTRL_Reset	Triggers a reset after specific data is cleared. This command can be used for hot start and cold start tests.
Get Receiver Version	BK_CTRL_GetVersion	Gets receiver version. The receiver will return version info in ACK message.
Set Receiver Clock Drift	BK_SETTIME_PresetPpb	Sets preset XO clock drift and drift range in unit of ppb.
Set Receiver Time	BK_AGNSS_Time_Inject	Injects time data from the host (MCU or computer).
Receiver Request for Assist Data	BK_AGNSS_Request	The receiver requests AGNSS data from the host.
NMEA Output Bitmask	BK_CTRL_SetNmeaBm BK_CTRL_ClrNmeaBm	Sets the output of NMEA messages without restarting the receiver. These commands do not change the settings of configuration parameters in the Flash memory. The settings of the configuration parameters in the Flash memory will be restored after the receiver is restarted.
Set Sensor Raw Data Output	BK_Sensor_Output	Sets IMU sensor raw data output

3.6.1 Deep Sleep

Table 3-7 BK_PWR_DeepSleep

Message	BK_PWR_DeepSleep			
Description	This command makes the receiver enter deep sleep mode to save power and set wakeup source.			
Firmware	-			
Comment	Supports two wake-up sources for deep sleep wake-up: • Event triggered (default): The GNSS receiver wakes up on receiving arbitrary data from UART. • Timer triggered: The GNSS receiver wakes up on a RTC timer time-out event.			
Message Structure	Header	MTYPE	STYPE	Length (Bytes)
	0x42 0x4B	0x00	0x03	8
Example	Event triggered (hex): 42 4b 51 05 00 03 00 08 00 00 00 00 00 00 00 00 Timer triggered (hex): 42 4b 73 23 00 03 00 08 80 00 00 3c 00 1e 00 00 (Work 60s; Hibernate for 60 seconds and save the configuration to the flash permanently)			

Payload description:

Byte Offset	Number Format	Scale	Name	Unit	Description

0	uint32	-	Wake	-	Wakeup source: • 0: Event triggered • 1: Timer triggered bit[0:15] of the parameter: indicates the 166X working time after the counter wakes up in s; bit[31] of the parameter: 1 indicates that the configuration of parameter 0 and parameter 1 will be saved in the flash, and the subsequent counter wake up is periodic, and no reconfiguration command is required.
4	uint32	-	TimerCnt	cycle	Cycles of 32k clock timer for wakeup, if set to timer triggered wakeup source. (32768 32K cycles for 1 second)

3.6.2 Stop Receiver

Table 3-8 BK_CTRL_Stop

<table>
<tr><td>Message</td><td colspan="4">BK_CTRL_Stop</td></tr>
<tr><td>Description</td><td colspan="4">Stops the operation of the receiver.
After sending this command, the receiver enters a completely stationary state and does not respond to any commands other than BK_CTRL_Start, and will continue to operate only when BK_CTRL_Start is received.</td></tr>
<tr><td>Firmware</td><td colspan="4">-</td></tr>
<tr><td>Comment</td><td colspan="4">-</td></tr>
<tr><td rowspan="2">Structure</td><td>Header</td><td>MTYPE</td><td>STYPE</td><td>LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x02</td><td>0x01</td><td>0</td></tr>
<tr><td>Example</td><td colspan="4">(Hex) 42 4b da 58 02 01 00 00</td></tr>
</table>

3.6.3 Start Receiver

Table 3-9 BK_CTRL_Start

Message	BK_CTRL_Start
Description	Starts the operation of the receiver (if it stopped operating)

© 2022-2024 Beken Corporation

	This command is used together with the BK_CTRL_Stop command. By default, the receiver will automatically start operating after power on, and there is no need to issue this command.			
Firmware	-			
Comment	-			
Structure	Header	MTYPE	STYPE	LEN (Bytes)
	0x42 0x4B	0x02	0x00	0
Example	(Hex) 42 4b ed 68 02 00 00 00			

3.6.4 Restart Receiver

Table 3-10 BK_CTRL_Reset

<table>
<tr><td>Message</td><td colspan="5">BK_CTRL_Reset</td></tr>
<tr><td>Description</td><td colspan="5">Triggers a watchdog reset</td></tr>
<tr><td>Firmware</td><td colspan="5">Version X166X22332 or later</td></tr>
<tr><td>Comment</td><td colspan="5">-</td></tr>
<tr><td rowspan="2">Message Structure</td><td>Header</td><td>MTYPE</td><td colspan="2">STYPE</td><td>LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x02</td><td colspan="2">0x05</td><td>8</td></tr>
<tr><td>Example</td><td colspan="5">Cold start, ClearFlag=1 (hex): 42 4b 1c 6b 02 05 00 08 00 00 00 01 00 00 00 01
Hot start, ClearFlag=0 (hex): 42 4b 0c 4a 02 05 00 08 00 00 00 01 00 00 00 00</td></tr>
<tr><td colspan="6">Payload description:</td></tr>
<tr><td>Byte Offset</td><td>Number Format</td><td>Scale</td><td>Name</td><td>Unit</td><td>Description</td></tr>
<tr><td>0</td><td>uint32</td><td>-</td><td>WaitTime</td><td>sec</td><td>Wait time (unit: second) (range: 1–30)</td></tr>
<tr><td>4</td><td>uint32</td><td>-</td><td>ClearFlag</td><td>-</td><td>• 1: Clear navigation data, cold start
• 0: Don’t clear navigation data, hot start</td></tr>
</table>

3.6.5 Get Receiver Version

Table 3-11 BK_CTRL_GetVersion

Message	BK_CTRL_GetVersion
Description	Gets receiver version

	The receiver will return version information in the ACK message.			
Firmware	-			
Comment	-			
Message Structure	Header	MTYPE	STYPE	LEN (Bytes)
	0x42 0x4B	0x02	0x07	0
Example	(Hex) 42 4b 68 f8 02 07 00 00			

3.6.6 Set Receiver Clock Drift

Table 3-12 BK_SETTIME_PresetPpb

Message	BK_SETTIME_PresetPpb			
Description	Sets preset XO clock drift and drift range in unit of ppb			
Firmware	-			
Comment	-			
Message Structure	Header	MTYPE	STYPE	LEN (Bytes)
	0x42 0x4B	0x18	0x02	8
Example	(Hex) 42 4b a5 b4 18 02 00 08 00 00 00 00 00 00 00 00			

Payload description:

Byte Offset	Number Format	Scale	Name	Unit	Description
0	INT32	-	Drift	ppb	Signed XO drift in ppb
4	uint32	-	Range	ppb	Unsigned XO drift range in ppb

3.6.7 Set Receiver Time

Table 3-13 BK_AGNSS_Time_Inject

Message	BK_AGNSS_Time_Inject
Description	The host (MCU or computer) injects the time data.
Firmware	-
Comment	Injects GPS System Time (GPST) BK defined message type, sent in little-endian order.

	Note: A leap second refers to a full-second adjustment before GPS and UTC.				
Message Structure	Header	MTYPE	STYPE	LEN (Bytes)	
	0x42 0x4B	0x18	0x03	12	
Payload description:					
Byte Offset	Number Format	Scale	Name	Unit	Description
0	uint32	1	Week	-	The number of full weeks corresponding to the GPS time
4	uint32	1	Msow	ms	The millisecond portion of the GPS time week
8	uint32	1	Leap	sec	The absolute value of leap seconds of UTC time

3.6.8 Receiver Request for Assist Data

Table 3-14 BK_AGNSS_Request

Message	BK_AGNSS_Request				
Description	The receiver requests AGNSS data from the host.				
Firmware	-				
Comment	BK defined message type, sent in little-endian order. The host can ignore the Lat_deg and Lon_deg data, provided that it can obtain the approximate position of the device and use it to collect ephemeris data of visible stars.				
Message Structure	Header	MTYPE	STYPE	LEN (Bytes)	
	0x42 0x4B	0x1C	0x00	12	
Example	(Hex) 42 4b b0 ab 1c 00 00 0c 00 00 00 0f 00 00 79 18 00 01 d8 a8				
Payload description:					
Byte Offset	Number Format	Scale	Name	Unit	Description
0	uint32	-	Bm_request	-	A bitmask showing the GNSS system that requests ephemeris: • Bit 0: GPS • Bit 1: BDS • Bit 2: GLO • Bit 3: GAL

4	INT32	1/1000	Lat_deg	deg	Latitude in degrees with a scale of 1/1000, i.e. latitude in 0.001 degrees (not used yet)
8	INT32	1/1000	Lon_deg	deg	Longitude in degrees with a scale of 1/1000, i.e. longitude in 0.001 degrees (not used yet)

3.6.9 NMEA Output Bitmask

Table 3-15 BK_CTRL_SetNmeaBm

<table>
<tr><td>Message</td><td colspan="4">BK_CTRL_SetNmeaBm</td></tr>
<tr><td>Description</td><td colspan="4">This command enables the output of NMEA messages without restarting the receiver. This command does not change the settings of user configuration parameters in the Flash memory. The settings of the user configuration parameters in the Flash memory will be restored after the receiver is restarted.</td></tr>
<tr><td>Firmware</td><td colspan="4">Version X166X2316X or later</td></tr>
<tr><td>Comment</td><td colspan="4">Output NMEA message mask. Bitmask, specifying which sentence is enabled for output. If any bit is set, the corresponding sentence(s) are enabled for output. If a bit is not set, there is no effect on the corresponding sentence. Each bit in the bitmask corresponds to a sentence:
• Bit 0: GGA
• Bit 1: GSA
• Bit 2: GSV
• Bit 3: VTG
• Bit 4: CNR
• Bit 5: RMC
• Bit 6: CLK
• Bit 7: POL
• Bit 8: THS
• Bit 9: ANT
• Bit 10: JAM
• Bit 11: INS
• Bit 12: GST
• Bit 13: GLL
• Bit 14: AID
• Bit 15: MSM
• Bit 16: GMP</td></tr>
<tr><td rowspan="2">Message Structure</td><td>Header</td><td>MTYPE</td><td>STYPE</td><td>LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x02</td><td>0x08</td><td>4</td></tr>
</table>

Example	(Hex) 42 4b 52 b1 02 08 00 04 00 00 00 3f

Table 3-16 BK_CTRL_ClrNmeaBm

<table>
<tr><td>Message</td><td colspan="4">BK_CTRL_ClrNmeaBm</td></tr>
<tr><td>Description</td><td colspan="4">This command disables the output of NMEA sentences without restarting the receiver. This command does not change the settings of user configuration parameters in the Flash memory. The settings of the user configuration parameters in the Flash memory will be restored after the receiver is restarted.</td></tr>
<tr><td>Firmware</td><td colspan="4">Version X166X2316X or later</td></tr>
<tr><td>Comment</td><td colspan="4">Output NMEA message mask. Bitmask, specifying which sentence is disabled for output. If any bit is set, the corresponding sentence(s) are disabled for output. If a bit is not set, there is no effect on the corresponding sentence. Each bit in the bitmask corresponds to a sentence:
Bit 0: GGA
Bit 1: GSA
Bit 2: GSV
Bit 3: VTG
Bit 4: CNR
Bit 5: RMC
Bit 6: CLK
Bit 7: POL
Bit 8: THS
Bit 9: ANT
Bit 10: JAM
Bit 11: INS
Bit 12: GST
Bit 13: GLL
Bit 14: AID
Bit 15: MSM
Bit 16: GMP</td></tr>
<tr><td rowspan="2">Message Structure</td><td>Header</td><td>MTYPE</td><td>STYPE</td><td>LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x02</td><td>0x0A</td><td>4</td></tr>
<tr><td>Example</td><td colspan="4">(Hex) 42 4b 52 b1 02 08 00 04 00 00 00 3f</td></tr>
</table>

3.6.10 IMU Sensor Raw Data Output

If the BK166X is connected to an external IMU sensor, the correct sensor model (IMU Device Type) and required sampling rate (IMU Data Rate) have been configured, and the Output Sensor Mode is configured to 'OUTPUT', the firmware can perform real-time IMU raw data collection and output.

Table 3-17 BK_Sensor_Output

Message	BK_Sensor_Output				
Description	Sets IMU sensor raw data output				
Firmware	VersionX166X23375 or later (16M)				
Comment	-				
Message Structure	Header	MTYPE	STYPE	LEN (Bytes)	
	0x42 0x4B	0x36	0x00	39	
Example	-				
Payload description:					
Byte Offset	Number Format	Scale	Name	Unit	Description
0	uint16	-	GPS_Week	-	GPS week number
2	double	-	GPS_Seconds	-	Seconds of GPS week
10	uint8	-	Sensor_Mask	-	A mask indicating which sensor data is valid: • 0x01: Time • 0x02: Gyroscope • 0x04: Accelerometer • 0x08: Temperature
14	float	-	Angular_Rate_X	deg/s	X-axis angular rate
18	float	-	Angular_Rate_Y	deg/s	Y-axis angular rate
22	float	-	Angular_Rate_Z	deg/s	Z-axis angular rate
26	float	-	Specific_Force_X	m/s^2	X-axis specific force
30	float	-	Specific_Force_Y	m/s^2	Y-axis specific force
34	float	-	Specific_Force_Z	m/s^2	Z-axis specific force
38	float	-	Temperature	°C	Temperature

3.7 GenCmdHex Tool Introduction

It is recommended to use the GenCmdHex tool to generate hexadecimal commands of the BK protocol, which can automatically implement the secondary conversion of specific commands, add CRC to the length calculation, and generate the secondary packet content.

GenCmdHex is generally included in the NavStarTool tool, and there is a GenCmdHex directory in the NavStarTool tool that contains the GenCmdHex executable file and the corresponding JSON configuration file.

The GenCmdHex tool divides messages into two categories: receiver configuration commands and general messages.

There are three main commands in the GenCmdHex interface: R, G, and L. R is used to generate commands related to receiver configuration (you need to enter the command Config ID and Parameter Value). G is used to generate general messages (you need to enter MTYPE, STYPE, the number of parameters, and the value of each parameter in sequence, and each parameter occupies 4 bytes). L is used to read and display the content of the GenCmdHex JSON file that saves the receiver configuration information.

After the Hex data is generated, any serial port tool can be used to send the command string to the receiver through UART0. For general commands, you can wait for the ACK message from the serial port. For receiver configuration commands, if there is an ACK message from the serial port and then the receiver restarts, it can be considered that the command has taken effect at this time.

Please note the following:

- Before using R to generate receiver configuration commands, you need to use L to read and display the contents of the GenCmdHex JSON file of the receiver information. The version of the GenCmdHex JSON file should be consistent with the format of the receiver's internal configuration parameters. This can be confirmed by checking and comparing the version information of the receiver’s internal configuration file with the "version" value in the GenCmdHex JSON file.
- When using the serial port tool to send the generated Hex data to the receiver, you need to confirm that the serial port baud rate and other settings match those of the receiver and that the serial port and the receiver can communicate correctly. Generally, the receiver will periodically output some information. If the configuration command to modify the baud rate is issued, the ACK will be sent at the original baud rate, and then the receiver will be restarted and apply the new baud rate.

4. RTCM Protocol

4.1 RTCM Introduction

The RTCM protocols are proposed by the Radio Technical Commission for Maritime Services to specify standards used in differential global positioning systems and real-time dynamic operations. The BK166X supports the output of GNSS reference station information, measurements, etc. that meet the RTCM standard RTCM 10403.3.

4.2 RTCM Frame Structure

The structure of a RTCM protocol message is illustrated in Figure 4-1.

Figure 4-1 RTCM Frame Structure

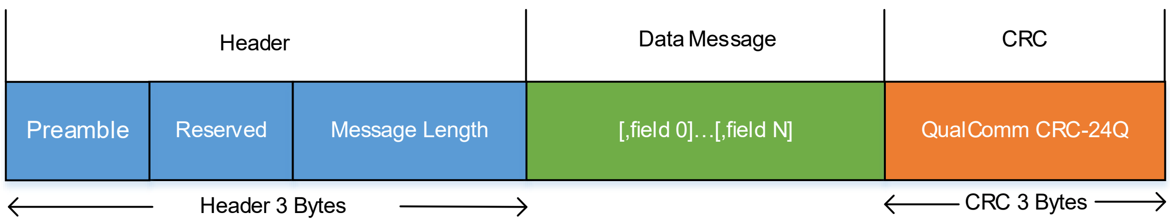


Table 4-1 RTCM Frame Structure

Field	No. of Bits	Description
Preamble	8	Set to '11010011' ('D3' in hex).
Reserved	6	Set to '000000'
Message Length	10	Message length in bytes
Data Message	Variable	Data message, 0–1023 bytes
CRC	24	QualComm CRC-24Q. The checksum is calculated starting and including the header up until, but excluding, the CRC field.

4.3 Supported RTCM Messages

Table 4-2 Supported RTCM Messages

Message	Message Type	Description
Reference station	1005, 1006	Reference station ID, antenna location, etc.

Message	Message Type	Description
Observations	1074, 1077, 1084, 1087, 1094, 1097, 1104, 1107, 1114, 1117, 1124, 1127	GNSS pseudoranges, carrier phases, Doppler, etc.
Ephemeris	1019, 1020, 1042, 1045, 1046	Ephemeris data

4.3.1 Reference Station Messages

Taking RTCM message type 1006 as an example, the contents for the reference station message are as shown in Table 4-3.

Table 4-3 Contents of RTCM Message Type 1006

Field No.	Data Field	Data Type	No. of Bits	Description
1	Message Number	uint12	12	Message number (“1006”= 0011 1110 1110)
2	Reference Station ID	uint12	12	Reference station ID
3	Reserved for ITRF realization year	uint6	6	The ITRF realization year identifies the datum definition used for coordinates in the message.
4	GPS Indicator	bit(1)	1	Whether GPS service is supported: • 1: Yes • 0: No
5	GLONASS Indicator	bit(1)	1	Whether GLONASS service is supported: • 1: Yes • 0: No
6	Galileo Indicator	bit(1)	1	Whether Galileo service is supported • 1: Yes • 0: No
7	Reference-Station Indicator	bit(1)	1	Reference-station indicator: • 1: Real, physical reference station • 0: Non-physical or computed reference station
8	Antenna Reference Point ECEF-X	int38	38	The antenna reference point X-coordinate is referenced to ITRF epoch as given in Field 3

© 2022-2024 Beken Corporation

Field No.	Data Field	Data Type	No. of Bits	Description
9	Single Receiver Oscillator Indicator	bit(1)	1	Single receiver oscillator indicator: • 1: All raw data observations are measured at the same instant • 0: All raw data observations may be measured at different instants.
10	Reserved	bit(1)	1	Reserved
11	Antenna Reference Point ECEF-Y	int38	38	The antenna reference point Y-coordinate is referenced to ITRF epoch as given in Field 3
12	Quarter Cycle Indicator	bit(2)	2	Carrier phase quarter cycle correction indicator: • 00: Correction status unspecified • 01: Corrected • 11: Uncorrected • 10: Reserved
13	Antenna Reference Point ECEF-Z	int38	38	The antenna reference point Z-coordinate is referenced to ITRF epoch as given in Field 3
14	Antenna Height	uint16	16	Antenna height

The contents of the type 1005 message are the same as those of the type 1006 message, except that 1006 also includes antenna height.

4.3.2 Observation Messages

The Multi Signal Message (MSM) is used to provide observation data for a variety of GNSS signals. The common MSMs are as shown in Table 4-4.

Table 4-4 RTCM MSM Messages

Message Type	MSM Type	Description
1074	MSM4	Full GPS pseudoranges and carrier phases plus signal strength
1077	MSM7	Full GPS pseudoranges, carrier phases, Doppler and signal strength (high resolution)
1084	MSM4	Full GLONASS pseudoranges and carrier phases plus signal strength

Message Type	MSM Type	Description
1087	MSM7	Full GLONASS pseudoranges, carrier phases, Doppler and signal strength (high resolution)
1094	MSM4	Full Galileo pseudoranges and carrier phases plus signal strength
1097	MSM7	Full Galileo pseudoranges, carrier phases, Doppler and signal strength (high resolution)
1104	MSM4	Full SBAS pseudoranges and carrier phases plus signal strength
1107	MSM7	Full SBAS pseudoranges, carrier phases, Doppler and signal strength (high resolution)
1114	MSM4	Full QZSS pseudoranges and carrier phases plus signal strength
1117	MSM7	Full QZSS pseudoranges, carrier phases, Doppler and signal strength (high resolution)
1124	MSM4	Full BeiDou pseudoranges and carrier phases plus signal strength
1127	MSM7	Full BeiDou pseudoranges, carrier phases, Doppler and signal strength (high resolution)

Note: The first three digits in the message type denote systems: 107 for GPS, 108 for GLONASS, 109 for Galileo, 110 for SBAS, 111 for QZSS, and 112 for BDS. The last digit represents MSM type: MSM4 and MSM7.

Each MSM message consists of three blocks as shown in Table 4-5.

Table 4-5 Content of an MSM Message

Block Type	Comment
Message header	Contains all information about satellites and signals that are transmitted in this message
Satellite data	Contains all satellite data that is common for all signals of a given satellite (e.g. Rough Range)
Signal data	Contains all signal data that is specific for each signal (e.g. Fine PhaseRange)

Taking MSM4 as an example, the contents of the above three blocks are shown in Table 4-6, Table 4-7, and Table 4-8.

Table 4-6 Content of MSM4 Message Header

Field No.	Data Field	Data Type	No. of Bits	Description
1	Message Number	uint12	12	Message type (e.g. 1074, 1084)
2	Reference Station ID	uint12	12	Reference station ID
3	GNSS Epoch Time	uint30	30	GNSS time, specific for each GNSS

© 2022-2024 Beken Corporation

Field No.	Data Field	Data Type	No. of Bits	Description
4	Multiple Message Bit	bit1(1)	1	MSM multiple message bit • 1: More MSMs follow for given physical time and reference station ID • 0: It is the last MSM for given physical time and reference station ID
5	IODS – Issue of Data Station	uint3	3	Issue of data station
6	Reserved	bit(7)	7	Reserved
7	Clock Steering Indicator	uint2	2	Clock steering indicator: • 0: Clock steering is not applied • 1: Clock steering has been applied • 2: Unknown clock steering status • 3: Reserved
8	External Clock Indicator	uint2	2	External clock indicator: • 0: Internal clock is used • 1: External clock is used, clock status is "locked" • 2: External clock is used, clock status is "not locked", • 3: Unknown clock is used
9	GNSS Divergence-free Smoothing Indicator	bit(1)	1	GNSS smoothing type indicator: • 1: Divergence-free smoothing is used • 0: Other type of smoothing is used
10	GNSS Smoothing Interval	bit(3)	3	GNSS smoothing interval
11	GNSS Satellite Mask	bit(64)	64	GNSS satellite mask If a bit is set to 1, it indicates that the corresponding GNSS satellite for which there is available data in this message. If a bit is set to 0, it indicates that the corresponding GNSS satellite for which there is no available data in this message.
12	GNSS Signal Mask	bit(32)	32	GNSS signal mask If a bit is set to 1, it indicates that the corresponding GNSS signal for which there is available data in this message. If a bit is set to 0, it indicates that the corresponding GNSS signal for which there is no available data in this message.

© 2022-2024 Beken Corporation

Field No.	Data Field	Data Type	No. of Bits	Description
13	GNSS Cell Mask	bit(X)	X (X≤64)	GNSS cell mask This field is of variable size: X=Nsat*Nsig, where Nsat is the number of satellites (the number of those bits set to 1 in Satellite Mask) and Nsig is the number of available signals (the number of those bits set to 1 in Signal Mask).

Table 4-7 Content of MSM4 Satellite Data

Field No.	Data Field	Data Type	No. of Bits	Description
1	The number of integer milliseconds in GNSS satellite rough ranges	uint8 (Nsat times)	8*Nsat	Rough range takes 18 bits which are split between two fields. This field contains the number of milliseconds in the satellite rough range.
2	GNSS Satellite rough ranges modulo 1 millisecond	uint10 (Nsat times)	10*Nsat	See the description above. Full rough range with accuracy 1/1024 milliseconds (about 300 m)

The content of MSM7 satellite data is the same as that of MSM4 satellite data, except that MSM7 also includes extended satellite information and PhaseRangeRates.

Table 4-8 Content of MSM4 Signal Data

Field No.	Data Field	Data Type	No. of Bits	Description
1	GNSS single fine Pseudoranges	int15 (Ncell times)	15*Ncell	GNSS signal fine pseudorange
2	GNSS single fine PhaseRange data	Int22 (Ncell times)	22*Ncell	GNSS signal fine phase range
3	GNSS PhaseRange Lock Time Indicator	uint4 (Ncell times)	4*Ncell	GNSS carrier phase lock time indicator The lock time indicator indicates whether the phase has a cycle slip, unlock, or other internal initialization status.
4	Half-cycle ambiguity indicator	bit1 (Ncell times)	1*Ncell	Half-cycle ambiguity indicator: • 0: No half-cycle ambiguity • 1: Half-cycle ambiguity
5	GNSS signal CNRs	int6 (Ncell times)	6* Ncell	GNSS signal carrier-to-noise ratio

Ncell refers to the number of signal data blocks, i.e. the number of bits set to 1 in the Cell Mask. Each data field is repeated Ncell times (internal looping is used). The order of the looped data corresponds to the order of bits in the Cell Mask.

The content of MSM7 signal data is the same as that of MSM4 signal data, except that MSM7 provides Pseudoranges with extended resolution, PhaseRange data with extended resolution, and CNRs with extended resolution, PhaseRange Lock Time Indicator with extended range and resolution, and also includes PhaseRangeRates.

5. Commonly Used Commands

This chapter lists commonly used receiver configuration commands. It should be noted that it is recommended to use the commands in Section 5.1 for configuration first to avoid abnormal configuration after modifying the Flash due to incorrect use of the configuration commands listed in Section 5.2.

5.1 Receiver Configuration Commands

5.1.1 Command Reply

Command Reply	PO_NMEA Command Reply	Comment
Command executed successfully	$OK	
Command execution failure	$FAIL	

If no message is returned to the command, check whether the content and format of the command are correct, or whether the communication with the chip is normal.

5.1.2 Version Number Inquiry

Function	PO_NMEA Command	Comment
Version number inquiry	$POLCFGPTVER	The version information is output with the $POLRS statement. For version information, see section 2.3.5

5.1.3 Set Serial Port Baud Rate

It becomes effective immediately. This configuration could be saved by 5.1.7 save to flash command.

Note that this baud rate is the normal baud rate output by customers. Currently, the baud rate in debug mode is fixed at a higher baud rate such as 2M due to the output of large amount of messages. Therefore, when the customer switches between the debug mode and the normal mode, the communication baud rate will be automatically switched without the need to increase or decrease the baud rate.

Function	PO_NMEA Command	Comment
9600	`$POLCFGPRT,9600`	

Function	PO_NMEA Command	Comment
38400	`$POLCFGPRT,38400`	
115200	`$POLCFGPRT,115200`	
230400	`$POLCFGPRT,230400`	
460800	`$POLCFGPRT,460800`	
1000000	`$POLCFGPRT,1000000`	
2000000	`$POLCFGPRT,2000000`	

5.1.4 TTFF Start Test

It becomes effective immediately.

Function	PO_NMEA Command	Comment
Host start	$POLCFGRESET,0	
Cold start	$POLCFGRESET,1	

5.1.5 Set Satellite Constellation Signals

It will become effective after 2 seconds. This configuration could be saved by 5.1.7 save to flash command.

Single-band users can configure the signals on the L1 frequency band:

Signal	PO_NMEA Command	Comment
GPS L1	$POLCFGSYS,1 or $POLCFGSYS,0001H	H represents hexadecimal input.
BDS B1I	$POLCFGSYS,4 or $POLCFGSYS,0004H	H represents hexadecimal input.
BDS B1I+B1C	$POLCFGSYS,36 or $POLCFGSYS,0024H	H represents hexadecimal input.
GLO G1	$POLCFGSYS,64 or $POLCFGSYS,0040H	H represents hexadecimal input.
GAL E1	$POLCFGSYS,128 or $POLCFGSYS,0080H	H represents hexadecimal input.
GPS L1+BDS B1I+B1C	$POLCFGSYS,229 or $POLCFGSYS,00E5H	H represents hexadecimal input.

Signal	PO_NMEA Command	Comment
+GLO G1+GAL E1		

Dual-band users can configure the signals on the L1 and L5 frequency bands:

Signal	PO_NMEA Command	Comment
GPS L1	$POLCFGSYS,1 or $POLCFGSYS,0001H	H represents hexadecimal input.
GPS L1+L5	$POLCFGSYS,3 or $POLCFGSYS,0003H	H represents hexadecimal input.
BDS B1I	$POLCFGSYS,4 or $POLCFGSYS,0004H	H represents hexadecimal input.
BDS B1I+B1C	$POLCFGSYS,36 or $POLCFGSYS,0024H	H represents hexadecimal input.
BDS B1I+B2A	$POLCFGSYS,12 or $POLCFGSYS,000CH	H represents hexadecimal input.
BDS B1I+B2A+B1C	$POLCFGSYS,44 or $POLCFGSYS,002CH	H represents hexadecimal input.
BDS B1I+B2A+B1C+B2I	$POLCFGSYS,16428 or $POLCFGSYS,402CH	H represents hexadecimal input.
GLO G1	$POLCFGSYS,64 or $POLCFGSYS,0040H	H represents hexadecimal input.
GAL E1	$POLCFGSYS,128 or $POLCFGSYS,0080H	H represents hexadecimal input.
GAL E1+E5A	$POLCFGSYS,384 or $POLCFGSYS,0180H	H represents hexadecimal input.
GPS L1+L5+IRS L5	$POLCFGSYS,8195 or $POLCFGSYS,2003H	H represents hexadecimal input.
GPS L1+L5 BDS B1I+B2A+B1C GLO G1 GAL E1+E5A IRS L5	$POLCFGSYS,8687 or $POLCFGSYS,21EFH	H represents hexadecimal input.

5.1.6 Set Beidou Modes

It will become effective after 2 seconds. This configuration could be saved by 5.1.7 save to flash command.

For 1661X series single-band chips, use the following command to configure:

Mode	PO_NMEA Command	Comment
BDS B1I+B1C	$POLCFGSYS,36 or $POLCFGSYS,0024H	H represents hexadecimal input.

© 2022-2024 Beken Corporation

For 1662X series dual-band chips, use the following commands to configure:

Mode	PO_NMEA Command	Comment
BDS B1I+B2A+B1C	$POLCFGSYS,44 or $POLCFGSYS,002CH	H represents hexadecimal input. This command is recommended for standard precision.
BDS B1I+B2A+B1C+B2I	$POLCFGSYS,16428 or $POLCFGSYS,402CH	H represents hexadecimal input. This command is recommended for RTK mode.

5.1.7 Set NMEA Message Output

It becomes effective immediately. This configuration could be saved by 5.1.7 save to flash command.

Function	PO_NMEA Command	Comment
Disable the output of all NMEA messages	`$POLCFGMSG,1,0`	
Enable the output of all NMEA messages	`$POLCFGMSG,1,1`	
Enable the output of GGA messages	`$POLCFGMSG,0,0,1`	
Enable the output of GSA messages	`$POLCFGMSG,0,1,1`	
Enable the output of GSV messages	`$POLCFGMSG,0,2,1`	
Enable the output of VTG messages	`$POLCFGMSG,0,3,1`	
Enable the output of CNR messages	`$POLCFGMSG,0,4,1`	
Enable the output of RMC messages	`$POLCFGMSG,0,5,1`	
Enable the output of CLK messages	`$POLCFGMSG,0,6,1`	
Enable the output of POL messages	`$POLCFGMSG,0,7,1`	
Enable the output of THS messages	`$POLCFGMSG,0,8,1`	
Enable the output of ANT messages	`$POLCFGMSG,0,9,1`	
Enable the output of JAM messages	`$POLCFGMSG,0,10,1`	
Enable the output of INS messages	`$POLCFGMSG,0,11,1`	

Function	PO_NMEA Command	Comment
Enable the output of GST messages	`$POLCFGMSG,0,12,1`	
Enable the output of GLL messages	`$POLCFGMSG,0,13,1`	
Enable the output of AID messages	`$POLCFGMSG,0,14,1`	
Enable the output of MSM messages	`$POLCFGMSG,0,15,1`	
Enable the output of GMP messages	`$POLCFGMSG,0,16,1`	
Enable the output of ODO messages	`$POLCFGMSG,0,17,1`	
Enable the output of ATT messages	`$POLCFGMSG,0,18,1`	
Enable the output of GFA messages	`$POLCFGMSG,0,19,1`	
Enable the output of ZDA messages	`$POLCFGMSG,0,20,1`	
Disable the output of GGA messages	`$POLCFGMSG,0,0,0`	
Disable the output of GSA messages	`$POLCFGMSG,0,1,0`	
Disable the output of GSV messages	`$POLCFGMSG,0,2,0`	
Disable the output of VTG messages	`$POLCFGMSG,0,3,0`	
Disable the output of CNR messages	`$POLCFGMSG,0,4,0`	
Disable the output of RMC messages	`$POLCFGMSG,0,5,0`	
Disable the output of CLK messages	`$POLCFGMSG,0,6,0`	
Disable the output of POL messages	`$POLCFGMSG,0,7,0`	
Disable the output of THS messages	`$POLCFGMSG,0,8,0`	
Disable the output of ANT messages	`$POLCFGMSG,0,9,0`	
Disable the output of JAM messages	`$POLCFGMSG,0,10,0`	
Disable the output of INS messages	`$POLCFGMSG,0,11,0`	
Disable the output of GST messages	`$POLCFGMSG,0,12,0`	
Disable the output of GLL messages	`$POLCFGMSG,0,13,0`	
Disable the output of AID messages	`$POLCFGMSG,0,14,0`	

Function	PO_NMEA Command	Comment
Disable the output of MSM messages	`$POLCFGMSG,0,15,0`	
Disable the output of GMP messages	`$POLCFGMSG,0,16,0`	
Disable the output of ODO messages	`$POLCFGMSG,0,17,0`	
Disable the output of ATT messages	`$POLCFGMSG,0,18,0`	
Disable the output of GFA messages	`$POLCFGMSG,0,19,0`	
Disable the output of ZDA messages	`$POLCFGMSG,0,20,0`	

5.1.8 RTCM Observation Output Rate

It becomes effective immediately. This configuration could be saved by 5.1.7 save to flash command.

Note: The second parameter shall be equal or less than receiver working rate or fix rate.

Function	PO_NMEA Command	Comment
Disable all RTCM messages	`$POLCFGRTCM,0,0,0`	Disable all RTCM messages discard MSM7 or MSM4 configuration.
Enable 1 Hz MSM7 RTCM message output	`$POLCFGRTCM,0,1,1`	
Enable 5 Hz MSM7 RTCM message output	`$POLCFGRTCM,0,5,1`	
Enable 10 Hz MSM7 RTCM message output	`$POLCFGRTCM,0,10,1`	
Enable 1 Hz MSM4 RTCM message output	`$POLCFGRTCM,1,1,1`	
Enable 5 Hz MSM4 RTCM message output	`$POLCFGRTCM,1,5,1`	
Enable 10 Hz MSM4 RTCM message output	`$POLCFGRTCM,1,10,1`	

5.1.9 Configuring the Mincnr value

Function	PO_NMEA Command	Comment
Configuring the Mincnr value	$POLCFGMCNR,30	Optional range: 5-40dB

5.1.10 Save Configuration to Flash

It becomes effective immediately.

Message format	$POLCFGSAVE
Example	$POLCFGSAVE
Description	This comment can be used to save current configuration.
Message type	Input

5.2 Binary Configuration Commands

After the receiver receives the commands in this section, the parameters will be saved to the internal RAM of the receiver. For some commands, the receiver will be restarted immediately after configuration, and the commands will not take effect until the receiver is restarted. It is recommended to use the commands in Section 5.1 and to avoid using the commands in this section.

If you need to modify a lot of configuration items, you should generate the configuration JSON file and then use download tools such as NavStartTool or other tools to directly update the chip configuration at one time.

Please note the following:

- The commands described in this section will be written directly into the Flash memory, so please use them with caution. It is generally recommended to use the commands in Section 5.1 instead of Section 5.2.
- Since the receiver will restart after executing some commands, the receiver will lose 1 to 2 seconds accordingly. After configuring the receiver, the user should wait for the receiver to start operating normally before performing subsequent operations.
- The configuration commands in this section only take effect at that time. If the receiver is powered off and restarted, the receiver will restore the configuration in the Flash memory. If you need to write this configuration to the Flash memory, please send the commands in this section to save the configuration to the Flash memory.

5.2.1 Command Reply

When a message is sent to the receiver, the receiver will send an ACK (acknowledgement) message back to the host, indicating the message has been received correctly.

The ACK message has the same format as the received message, except that its MTYPE value will be increased by 1 based on the received message. Its STYPE value is the same as the received message, and its data segment length may be empty.

If no message is returned to the command, check whether the content and format of the command are correct, or whether the communication with the chip is normal.

5.2.2 Version Number Inquiry

Function	Command (Hex)	Comment
Version number inquiry	42 4b 68 f8 02 07 00 00	The version number is output with the $POLRS statement.

5.2.3 Chip Information Inquiry

Function	Command (Hex)	Comment
Chip Information inquiry	42 4b af 39 02 0d 00 00	The chip information is output with the $BKCHIP statement. Supports X166X24365 and later versions.

5.2.4 TTFF Start Test

Function	Command (Hex)	Comment
Hot start	42 4b 54 cb 02 0c 00 04 00 00 00 00	
Cold start	42 4b 4a 3b 02 0c 00 04 00 00 00 ff	
Factory reset	42 4b 49 c4 02 0c 00 04 00 00 ff ff	

5.2.5 Set Operating Frequency Band

Please note the following:

© 2022-2024 Beken Corporation

- After changing the operating frequency band configuration, you need to save the configuration to Flash, and the configuration will become effective after the chip is powered off and restarted.
- For dual-band mode to single-band mode, the L5 signals are restricted after restarting. Currently, only the L1 frequency band works.
- For single-band mode to dual-band mode, you need to configure and add L5 signals yourself.

Function	Command (Hex)	Comment
Single band	42 4b e8 85 26 02 00 0c 55 55 00 01 00 00 00 01 00 00 00 01	
Dual band	42 4b d8 e6 26 02 00 0c 55 55 00 01 00 00 00 01 00 00 00 02	

5.2.6 Set Core Voltage

Note: After changing the working voltage configuration, you need to save the configuration to Flash, and the configuration will become effective after the chip is powered off and restarted.

Function	Command (Hex)	Comment
0.8 V	42 4b 97 7e 26 02 00 0c 55 55 00 01 00 00 00 02 00 00 00 08	Only in the case of 1 Hz NMEA output and single band, the core voltage can be configured to 0.8 V, and the performance will be slightly reduced.
0.9 V	42 4b 87 5f 26 02 00 0c 55 55 00 01 00 00 00 02 00 00 00 09	

5.2.7 Set Serial Port Baud Rate

Note that this baud rate is the normal baud rate output by customers. Currently, the baud rate in debug mode is fixed at a higher baud rate such as 2M due to the large amount of message output. Therefore, when the customer switches between the debug mode and the normal mode, the communication baud rate will be automatically switched without the need to increase or decrease the baud rate.

Note: For 23A355 and previous versions, the configuration needs to be saved to Flash, and it will become effective after the chip is powered off and restarted. For 23A365 and later versions, the configuration becomes effective immediately without a restart of the receiver, and it can be saved to Flash through commands.

Function	Command (Hex)	Comment
9600	42 4b b3 68 26 02 00 0c 55 55 00 01 00 00 00 04 00 00 25 80	

Function	Command (Hex)	Comment
38400	42 4b 69 be 26 02 00 0c 55 55 00 01 00 00 00 04 00 00 96 00	
115200	42 4b 9c f5 26 02 00 0c 55 55 00 01 00 00 00 04 00 01 c2 00	
230400	42 4b 55 ff 26 02 00 0c 55 55 00 01 00 00 00 04 00 03 84 00	
460800	42 4b d7 ca 26 02 00 0c 55 55 00 01 00 00 00 04 00 07 08 00	
1000000	42 4b d4 a8 26 02 00 0c 55 55 00 01 00 00 00 04 00 0f 42 40	
2000000	42 4b c5 45 26 02 00 0c 55 55 00 01 00 00 00 04 00 1e 84 80	

5.2.8 Set Positioning Solution Rate

The setting changes the positioning solution rate, and also changes the output rate of NMEA positioning messages.

Function	Command (Hex)	Comment
1 Hz	42 4b ac 06 26 02 00 0c 55 55 00 01 00 00 00 03 00 00 00 01	
2 Hz	42 4b 9c 65 26 02 00 0c 55 55 00 01 00 00 00 03 00 00 00 02	
5 Hz	42 4b ec 82 26 02 00 0c 55 55 00 01 00 00 00 03 00 00 00 05	
10 Hz	42 4b 1d 6d 26 02 00 0c 55 55 00 01 00 00 00 03 00 00 00 0a	
20 Hz	42 4b ee 92 26 02 00 0c 55 55 00 01 00 00 00 03 00 00 00 14	

5.2.9 Set Satellite Constellation Signals

The following commands can enable or disable the satellite signal. For example, sending the first GPSL1 enable command listed below will enable the GPSL1 signal based on the existing system signals. If the GPSL1 signal is already enabled, this command will be ignored. If you need to configure multi-signal mixed positioning, you can send the corresponding signal enable commands in sequence to enable the signals.

Note: Set the QZSS system signals using the GPS system enable/disable commands.

Function	Command (Hex)	Comment
Enable GPSL1	42 4b 61 83 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 01	
Enable GPSL5	42 4b 51 e0 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 02	

© 2022-2024 Beken Corporation

Function	Command (Hex)	Comment
Enable BDSB1I	42 4b 31 26 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 04	
Enable BDSB2A	42 4b f0 aa 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 08	
Enable BDSB2B	42 4b 63 93 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 10	
Enable BDSB1C	42 4b 55 c0 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 20	
Enable GLOG1	42 4b 39 66 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 40	
Enable GALE1	42 4b e0 2a 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 00 80	
Enable GALE5A	42 4b 42 93 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 01 00	
Enable GALE5B	42 4b 17 c0 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 02 00	
Enable IRSL5	42 4b 77 44 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 20 00	
Enable BD2B2I	42 4b 7c 6e 26 02 00 0c 55 55 00 01 00 00 00 05 00 00 40 00	
Enable SBASL1	42 4b 21 d6 26 02 00 0c 55 00 24 41 00 00 00 05 00 00 10 00	
Disable GPSL1	42 4b 19 14 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 01	
Disable GPSL5	42 4b 29 77 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 02	
Disable BDSB1I	42 4b 49 b1 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 04	
Disable BDSB2A	42 4b 88 3d 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 08	
Disable BDSB2B	42 4b 1b 04 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 10	
Disable BDSB1C	42 4b 2d 57 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 20	
Disable GLOG1	42 4b 41 f1 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 40	
Disable GALE1	42 4b 98 bd 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 00 80	
Disable GALE5A	42 4b 3a 04 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 01 00	
Disable GALE5B	42 4b 6f 57 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 02 00	
Disable IRSL5	42 4b 0f d3 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 20 00	
Disable BD2B2I	42 4b 04 f9 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 40 00	
Disable SBASL1	42 4b 0a 46 26 05 00 0c 55 55 00 01 00 00 00 05 00 00 10 00	

© 2022-2024 Beken Corporation

5.2.10 Set Reference Time System for Observation Output

Function	Command (Hex)	Comment
GPST	42 4b 50 d8 26 02 00 0c 55 55 00 01 00 00 00 08 00 00 00 00	
BDST	42 4b 40 f9 26 02 00 0c 55 55 00 01 00 00 00 08 00 00 00 01	
GLOT	42 4b 70 9a 26 02 00 0c 55 55 00 01 00 00 00 08 00 00 00 02	
GALT	42 4b 60 bb 26 02 00 0c 55 55 00 01 00 00 00 08 00 00 00 03	

5.2.11 Set Reference Time System for Receiver

Function	Command (Hex)	Comment
GPST	42 4b fa 89 26 02 00 0c 55 55 00 01 00 00 00 09 00 00 00 00	
BDST	42 4b ea a8 26 02 00 0c 55 55 00 01 00 00 00 09 00 00 00 01	
GLOT	42 4b da cb 26 02 00 0c 55 55 00 01 00 00 00 09 00 00 00 02	
GALT	42 4b ca ea 26 02 00 0c 55 55 00 01 00 00 00 09 00 00 00 03	

5.2.12 Set Reference Time System for PPS

Function	Command (Hex)	Comment
GPST	42 4b 14 5b 26 02 00 0c 55 55 00 01 00 00 00 0a 00 00 00 00	
BDST	42 4b 04 7a 26 02 00 0c 55 55 00 01 00 00 00 0a 00 00 00 01	
GLOT	42 4b 34 19 26 02 00 0c 55 55 00 01 00 00 00 0a 00 00 00 02	
GALT	42 4b 24 38 26 02 00 0c 55 55 00 01 00 00 00 0a 00 00 00 03	

5.2.13 Set NMEA Message Output

The following commands become effective immediately after configuration. After the receiver is restarted or shut down, it will restore to the NMEA sentence output setting in Flash.

Note: Do not configure the baud rate too low, otherwise it will cause communication abnormalities.

The following commands are supported in X166X2316X and later versions.

© 2022-2024 Beken Corporation

Function	Command (Hex)	Comment
Disable the output of all NMEA messages	42 4b 6c 21 02 0a 00 04 ff ff ff ff	
Enable the output of all NMEA messages	42 4b 0c c2 02 08 00 04 ff ff ff ff	
Enable the output of GGA messages	42 4b 85 2c 02 08 00 04 00 00 00 01	
Enable the output of GSA messages	42 4b b5 4f 02 08 00 04 00 00 00 02	
Enable the output of GSV messages	42 4b d5 89 02 08 00 04 00 00 00 04	
Enable the output of VTG messages	42 4b 14 05 02 08 00 04 00 00 00 08	
Enable the output of CNR messages	42 4b 87 3c 02 08 00 04 00 00 00 10	
Enable the output of RMC messages	42 4b b1 6f 02 08 00 04 00 00 00 20	
Enable the output of CLK messages	42 4b dd c9 02 08 00 04 00 00 00 40	
Enable the output of POL messages	42 4b 04 85 02 08 00 04 00 00 00 80	
Enable the output of THS messages	42 4b a6 3c 02 08 00 04 00 00 01 00	
Enable the output of ANT messages	42 4b f3 6f 02 08 00 04 00 00 02 00	
Enable the output of JAM messages	42 4b 59 c9 02 08 00 04 00 00 04 00	
Enable the output of INS messages	42 4b 1c a4 02 08 00 04 00 00 08 00	
Enable the output of GST messages	42 4b 96 7e 02 08 00 04 00 00 10 00	

Function	Command (Hex)	Comment
Enable the output of GLL messages	42 4b 93 eb 02 08 00 04 00 00 20 00	
Enable the output of AID messages	42 4b 98 c1 02 08 00 04 00 00 40 00	
Enable the output of MSM messages	42 4b 8e 95 02 08 00 04 00 00 80 00	
Enable the output of GMP messages	42 4b a2 3d 02 08 00 04 00 01 00 00	
Enable the output of ODO messages	42 4b fb 6d 02 08 00 04 00 02 00 00	
Enable the output of ATT messages	42 4b 49 cd 02 08 00 04 00 04 00 00	
Enable the output of GFA messages	42 4b 3c ac 02 08 00 04 00 08 00 00	
Enable the output of ZDA messages	42 4b d6 6e 02 08 00 04 00 10 00 00	
Disable the output of a single NMEA message		
Disable the output of a single GGA message	42 4b e5 cf 02 0a 00 04 00 00 00 01	
Disable the output of a single GSA message	42 4b d5 ac 02 0a 00 04 00 00 00 02	
Disable the output of a single GSV message	42 4b b5 6a 02 0a 00 04 00 00 00 04	
Disable the output of a single VTG message	42 4b 74 e6 02 0a 00 04 00 00 00 08	
Disable the output of a single CNR message	42 4b e7 df 02 0a 00 04 00 00 00 10	
Disable the output of a single RMC message	42 4b d1 8c 02 0a 00 04 00 00 00 20	

Function	Command (Hex)	Comment
Disable the output of a single CLK message	42 4b bd 2a 02 0a 00 04 00 00 00 40	
Disable the output of a single POL message	42 4b 64 66 02 0a 00 04 00 00 00 80	
Disable the output of a single THS message	42 4b c6 df 02 0a 00 04 00 00 01 00	
Disable the output of a single ANT message	42 4b 93 8c 02 0a 00 04 00 00 02 00	
Disable the output of a single JAM message	42 4b 39 2a 02 0a 00 04 00 00 04 00	
Disable the output of a single INS message	42 4b 7c 47 02 0a 00 04 00 00 08 00	
Disable the output of a single GST message	42 4b f6 9d 02 0a 00 04 00 00 10 00	
Disable the output of a single GLL message	42 4b f3 08 02 0a 00 04 00 00 20 00	
Disable the output of a single AID message	42 4b f8 22 02 0a 00 04 00 00 40 00	
Disable the output of a single MSM message	42 4b ee 76 02 0a 00 04 00 00 80 00	
Disable the output of a single GMP message	42 4b c2 de 02 0a 00 04 00 01 00 00	
Disable the output of a single ODO message	42 4b 9b 8e 02 0a 00 04 00 02 00 00	
Disable the output of a single ATT message	42 4b 29 2e 02 0a 00 04 00 04 00 00	
Disable the output of a single GFA message	42 4b 5c 4f 02 0a 00 04 00 08 00 00	
Disable the output of a single ZDA message	42 4b b6 8d 02 0a 00 04 00 10 00 00	

© 2022-2024 Beken Corporation

5.2.14 Set the MSM Type for RTCM MSM Message Output

Function	Command (Hex)	Comment
MSM7	42 4b 56 af 26 02 00 0c 55 55 00 01 00 00 00 10 00 00 00 00	
MSM5	42 4b d5 a0 26 02 00 0c 55 00 24 81 00 00 00 10 00 00 00 03	
MSM4	42 4b 46 8e 26 02 00 0c 55 55 00 01 00 00 00 10 00 00 00 01	
MSM3	42 4b a3 75 26 02 00 0c 55 00 24 32 00 00 00 10 00 00 00 02	

5.2.15 Set Positioning Solution Mode

Function	Command (Hex)	Comment
SPS	42 4b c9 ff 26 02 00 0c 55 55 00 01 00 00 00 0c 00 00 00 01	
SBS	42 4b f9 9c 26 02 00 0c 55 55 00 01 00 00 00 0c 00 00 00 02	
RTD	42 4b e9 bd 26 02 00 0c 55 55 00 01 00 00 00 0c 00 00 00 03	
RTK	42 4b 99 5a 26 02 00 0c 55 55 00 01 00 00 00 0c 00 00 00 04	
DUALANT	42 4b 12 c6 26 02 00 0c 55 00 23 71 00 00 00 0c 00 00 00 07	

5.2.16 Set Receiver Operating Modes

Function	Command (Hex)	Comment
Auto	42 4b 73 8f 26 02 00 0c 55 55 00 01 00 00 00 0d 00 00 00 00	
Static	42 4b 63 ae 26 02 00 0c 55 55 00 01 00 00 00 0d 00 00 00 01	
Flight	42 4b f2 87 26 02 00 0c 55 55 00 01 00 00 00 0d 00 00 00 08	
Badge	42 4b 59 59 26 02 00 0c 55 00 23 71 00 00 00 0d 00 00 00 09	
Wearable	42 4b 69 3a 26 02 00 0c 55 00 23 71 00 00 00 0d 00 00 00 0a	

© 2022-2024 Beken Corporation

5.2.17 Enable/Disable Static Detection

Function	Command (Hex)	Comment
Enable	42 4b fc fe 26 02 00 0c 55 55 00 01 00 00 00 11 00 00 00 00	
Disable	42 4b ec df 26 02 00 0c 55 55 00 01 00 00 00 11 00 00 00 01	

5.2.18 Set RTCM Observation Output Rate

Function	Command (Hex)	Comment
1 Hz	42 4b 25 00 26 02 00 0c 55 55 00 01 00 00 00 07 00 00 00 01	
2 Hz	42 4b 15 63 26 02 00 0c 55 55 00 01 00 00 00 07 00 00 00 02	
5 Hz	42 4b 65 84 26 02 00 0c 55 55 00 01 00 00 00 07 00 00 00 05	
10 Hz	42 4b 94 6b 26 02 00 0c 55 55 00 01 00 00 00 07 00 00 00 0a	
20 Hz	42 4b 67 94 26 02 00 0c 55 55 00 01 00 00 00 07 00 00 00 14	

5.2.19 Set RTCM Observation Output

Function	Command (Hex)	Comment
Disable all observation outputs	42 4b 4f b5 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 00	
Enable all observation outputs	42 4b 51 45 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 ff	
Enable MEAS output	42 4b 5f 94 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 01	
Enable EPH output	42 4b 6f f7 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 02	
Enable ION output	42 4b 0f 31 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 04	
Enable SBS output	42 4b ce bd 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 08	
Enable GEO output	42 4b 5d 84 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 10	
Enable MEAS+ EPH output	42 4b 7f d6 26 02 00 0c 55 00 23 71 00 00 00 31 00 00 00 03	

© 2022-2024 Beken Corporation

5.2.20 Save Configuration to Flash

Note: Some configurations will become effective after the receiver is powered off and restarted.

Function	Command (Hex)	Comment
Save configuration to Flash	42 4b cc 17 26 04 00 00	

5.2.21 Setting Nmea Decimal Digit

Function	Command (Hex)	Comment
Latitude and longitude digit 4	42 4b 7c fa 26 02 00 0c 55 00 24 44 00 00 00 3a 00 00 00 04	
Latitude and longitude digit 5	42 4b 6c db 26 02 00 0c 55 00 24 44 00 00 00 3a 00 00 00 05	
Latitude and longitude digit 6	42 4b 5c b8 26 02 00 0c 55 00 24 44 00 00 00 3a 00 00 00 06	
Latitude and longitude digit 7	42 4b 4c 99 26 02 00 0c 55 00 24 44 00 00 00 3a 00 00 00 07	
Orthometric height digit 1	42 4b d9 7c 26 02 00 0c 55 00 24 44 00 00 00 51 00 00 00 01	
Orthometric height digit 2	42 4b e9 1f 26 02 00 0c 55 00 24 44 00 00 00 51 00 00 00 02	
Orthometric height digit 3	42 4b f9 3e 26 02 00 0c 55 00 24 44 00 00 00 51 00 00 00 03	
Orthometric height digit 4	42 4b 89 d9 26 02 00 0c 55 00 24 44 00 00 00 51 00 00 00 04	
UTC second digit 2	42 4b 07 cd 26 02 00 0c 55 00 24 44 00 00 00 52 00 00 00 02	
UTC second digit 3	42 4b 17 ec 26 02 00 0c 55 00 24 44 00 00 00 52 00 00 00 03	
DOP digit 1	42 4b 9d ff 26 02 00 0c 55 00 24 44 00 00 00 53 00 00 00 01	
DOP digit 2	42 4b ad 9c 26 02 00 0c 55 00 24 44 00 00 00 53 00 00 00 02	

Velocity digit 1	42 4b fa 2b 26 02 00 0c 55 00 24 44 00 00 00 54 00 00 00 01	
Velocity digit 2	42 4b ca 48 26 02 00 0c 55 00 24 44 00 00 00 54 00 00 00 02	
Velocity digit 3	42 4b da 69 26 02 00 0c 55 00 24 44 00 00 00 54 00 00 00 03	

5.2.22 Setting Trigger Time Information

Function	Command (Hex)	Comment
Enable the output of HPTS messages	42 4b 4c 7d 26 02 00 0c 55 00 24 33 00 00 00 4d 00 00 00 01	
Disable the output of HPTS message	42 4b 5c 5c 26 02 00 0c 55 00 24 33 00 00 00 4d 00 00 00 00	

6. AGNSS

AGNSS service can be divided into Online and Offline (Extended ephemeris) service, extended ephemeris support up to 7 ephemeris forecast.

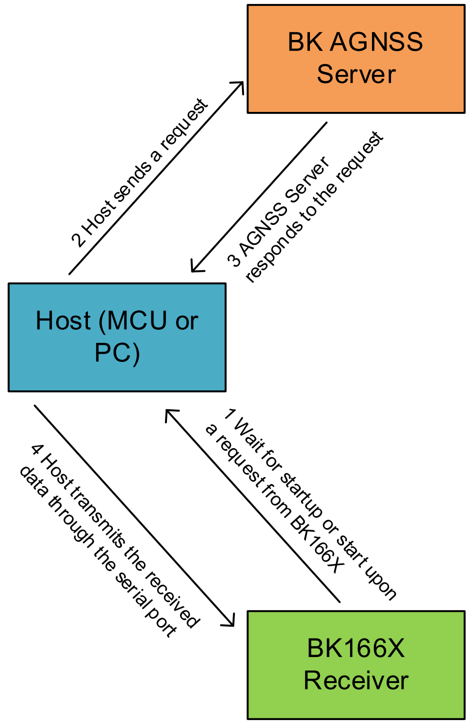
6.1 AGNSS Workflow

The host is mainly responsible for obtaining and issuing AGNSS ephemeris and AGNSS time. The ephemeris data is encoded in RTCM 3. X format, and the time data is encoded in BK protocol format. The basic workflow is as follows:

1. The host can start the AGNSS session when there is a network connection after confirming that the BK166X receiver starts outputting information normally, or after an AGNSS request from the BK166X receiver is received.
2. The host sends an MQTT subscription request to the BK AGNSS Server through the MQTT protocol.
3. BK AGNSS Server responds to the subscription request from the host and sends AGNSS ephemeris and AGNSS time data to the host.
4. When the host obtains AGNSS ephemeris and AGNSS time data, it can immediately send it to the BK166X receiver or wait for all data to be received completely before sending the data to the BK166X receiver. The received data can be directly transmitted to the BK166X receiver through the serial port.

Note: Due to the large amount of extended ephemeris data, do not inject all data at once. It is recommended to inject a maximum of 3k bytes of data per second, and at least 1k bytes of data; The real-time ephemeris needs to be injected as soon as possible, and if the injection time is too long, it will delay the first positioning time.

Figure 6-1 AGNSS Workflow



6.2 AGNSS Service

The AGNSS service uses the MQTT to send ephemeris and time messages. The relevant information of the MQTT service is as follows:

- Address: 43.154.207.218
- Port: 1883
- Username: ******
- Password: ******

Please contact customer support to provide the username and password.

6.3 MQTT Client Message Subscription

There are two types of messages that require subscription to the ANGSS service, namely ephemeris and time messages:

- Topic of time: /TIME
- Topic of time data transparent transmission: /TIMEP
- Topic of Real-time ephemeris: /EPH/X_Y/SYS/PRN

- Topic of Extended ephemeris: /EEPH[1/3]/X_Y/SYS/PRN

Where:

The brackets [1/3] indicate optional options.

For example, /EEPH/X_Y is used, the system will subscribe to the extended ephemeris with a forecast length of 7 days by default.

When option 1 is added, such as using "/EEPH1/X_Y", you can subscribe to the extended ephemeris of 1day.

When optional 3 is added, such as using "/EEPH3/X_Y", you can subscribe to the extended ephemeris of 3 days.

- X and Y represent the user's region, and the calculation method is as follows:

  X=INT((80–latitude)/5), 0≤X<32

  Latitude range: -80°<latitude<80°, north latitude is positive, south latitude is negative; latitude≥80°, X=0, latitude≤-80°, X=31

  Y=INT((longitude+180)/5), 0≤Y<72

  Longitude range: -180°≤longitude<180°, east longitude is positive, west longitude is negative

- SYS represents the satellite system, including GPS, BDS, GLO, and GAL.
- PRN represents the satellite PRN number.

For example:

Chinese domestic users request ephemerides of visible stars for all systems:

- System-wide: /EPH/9_58/#
- Single GPS: /EPH/9_58/GPS/#
- System-wide extension ephemeris of 7 days: /EEPH/9_58/#
- System-wide extension ephemeris of 3 days:: /EEPH3/9_58/#
- ……..

Russian users (based on Moscow latitude and longitude) request ephemerides of visible stars for all systems:

- System-wide: /EPH/5_43/#
- Single GPS: /EPH/5_43/GPS/#
- System-wide extension ephemeris of 7 days: /EEPH/5_43/#`
- System-wide extension ephemeris of 3 days:: /EEPH3/5_43/#
- ……..

Please note the following:

1. The ephemeris is the same for single band and dual band. Single band and dual band can be injected into the ephemeris in the same way.
2. The speed at which the server issues ephemeris is related to the network speed of the host. The speed at which the server issues ephemeris for a single system is the same as for multiple systems.
3. The response speed of the chip's injection of ephemeris is related to the communication bandwidth between the host and the chip. There is almost no difference in the chip's response speed between a single system and multiple systems.

6.4 Host AGNSS Function Implementation

The host needs to implement the MQTT client protocol, which is a lightweight IoT communication protocol with multiple open-source implementations. Before implementing relevant functions, it is necessary to ensure that MQTT communication transmission is effective. Normally, the data size for each subscription request is within 4KB.

The host needs to perform serial communication. You need to confirm that the baud rate of the serial port is correct, and the up and down paths of the serial port are in a proper state, which can be confirmed by sending a version query or a temporary command to modify the NMEA message output.

The example source code for AGNSS implementation on the host is as follows:

```
#include <QcoreApplication>
#include <QtDebug>
#include "BkMqttService.h"
int main(int argc, char *argv[])
{
    QcoreApplication a(argc, argv);
    const Qstring 101assword("agnssClient01");
    const Qstring 101assword("AgnssClient01!");
    const Qstring serverUrl("114.119.119.26");
    const quint16 port = 1883;
    const qreal latInDeg{31.2302}, lonInDeg{121.1203};
    BkMqttService *service = new BkMqttService(101assword, 101assword, serverUrl, port,
&a);
    Qobject::connect(service, &BkMqttService::ephMessageReceived, [](QbyteArray data)
    {
        qInfo() << "receiving eph, data size " << data.size();
    });
    Qobject::connect(service, &BkMqttService::timeMessageReceived, [](QbyteArray data)
    {
        qInfo() << "receiving time, data size " << data.size();
    });
    Qobject::connect(service, &BkMqttService::aboutToClose, &a,
&QcoreApplication::quit);
```

```
    service->start(latInDeg, lonInDeg, 0Xff);
    return a.exec();
}
```

```
void BkMqttService::onSubscribed(const Qstring topic, quint8 qos)
{
    QMQTT::Client *client = qobject_cast<QMQTT::Client*>(sender());

    if (client != Q_NULLPTR)
    {
        information(Qstring("%1 have subscribed %2,qos %3").arg(client->clientId(),
topic).arg(qos));
    }
}
```

```
QstringList BkMqttService::topics(qreal lat, qreal lon) const
{
    QstringList topicList;
    const Qpair<qint32, qint32> indexOfLocaton = gridIndex(lat, lon);
    if (indexOfLocaton.first == kInvalidIndex)
    {
        information(Qstring("invalid AGNSS position lat %1,lon %2").arg(lat).arg(lon));
        return topicList;
    }
    topicList << "/TIME";
    Qstring ephTopicPrefix = Qstring("/EPH/%1_%2/").arg(indexOfLocaton.first)
                                                    .arg(indexOfLocaton.second);
    QstringList gnssNameList = sysList();
    for (int i = 0; i < sysList().size(); ++i)
    {
        if (((mSysMask >> i) & 0x01) == 0x01)
        {
            Qstring topic(ephTopicPrefix);
            topic.append(sysList().at(i));
            topicList << topic;
        }
    }
    return topicList;
}
```

```
constexpr qint32 kInvalidIndex{-180};
Qpair<qint32, qint32> BkMqttService::gridIndex(qreal lat, qreal lon)
{
    constexpr qreal kMaxLat{80.0}, kMinLat{-80}, kMinLon{-180.0}, kGridStep{5.0};
    if (lat > kMaxLat || lat < kMinLat)
    {
        return {kInvalidIndex, kInvalidIndex};
    }
    qint32 latIndex = static_cast<qint32>((kMaxLat - lat) / kGridStep);
    qint32 lonIndex = static_cast<qint32>((lon - kMinLon) / kGridStep);
    return {latIndex, lonIndex};
}
```

6.5 Issue of AGNSS Time

For the format of AGNSS time message issued by the host, please refer to Section 3.5.7 Set Receiver Time, Or directly use the "/TIMEP" theme AGNSS time information transparent transmission to set the receiver time.

Note: If you do not need to develop the AGNSS server, you do not need to pay attention to this section.

6.6 Issue of AGNSS Ephemeris

After successfully sending AGNSS time messages, the host can issue ephemeris messages. For ephemeris data of different systems please refer to the corresponding RTCM messages:

- Please refer to RTCM protocol message type 1019 for GPS ephemeris.
- Please refer to RTCM protocol message type 1020 for GLONASS ephemeris.
- Please refer to RTCM protocol message type 1042 for BDS ephemeris.
- Please refer to RTCM protocol message type 1045/1046 for GALILEO ephemeris.

The Extended Ephemeris uses the custom RTCM protocol, where:

- Please refer to RTCM protocol message type 1919 for GPS Extended Ephemeris.
- Please refer to RTCM protocol message type 1920 for GLONASS Extended Ephemeris.
- Please refer to RTCM protocol message type 1942 for BDS Extended Ephemeris.
- Please refer to RTCM protocol message type 1946 for GALILEO Extended Ephemeris.

Note: Due to the limited memory size of the chip, the host only needs to send the ephemeris corresponding to the X_Y region (see Section 6.3 for X_Y calculation), otherwise the positioning performance of the chip may be affected.

The total number of ephemeris messages sent by the host within three seconds: GPS+QZSS ephemeris≤24, BDS ephemeris ≤26, GLO ephemeris≤14, and GAL ephemeris≤16.

Note: If you do not need to develop the AGNSS server, you do not need to pay attention to this section.

6.7 AGNSS Ephemeris Collection Situation

6.7.1 Real-time AGNSS Ephemeris Collection Situation

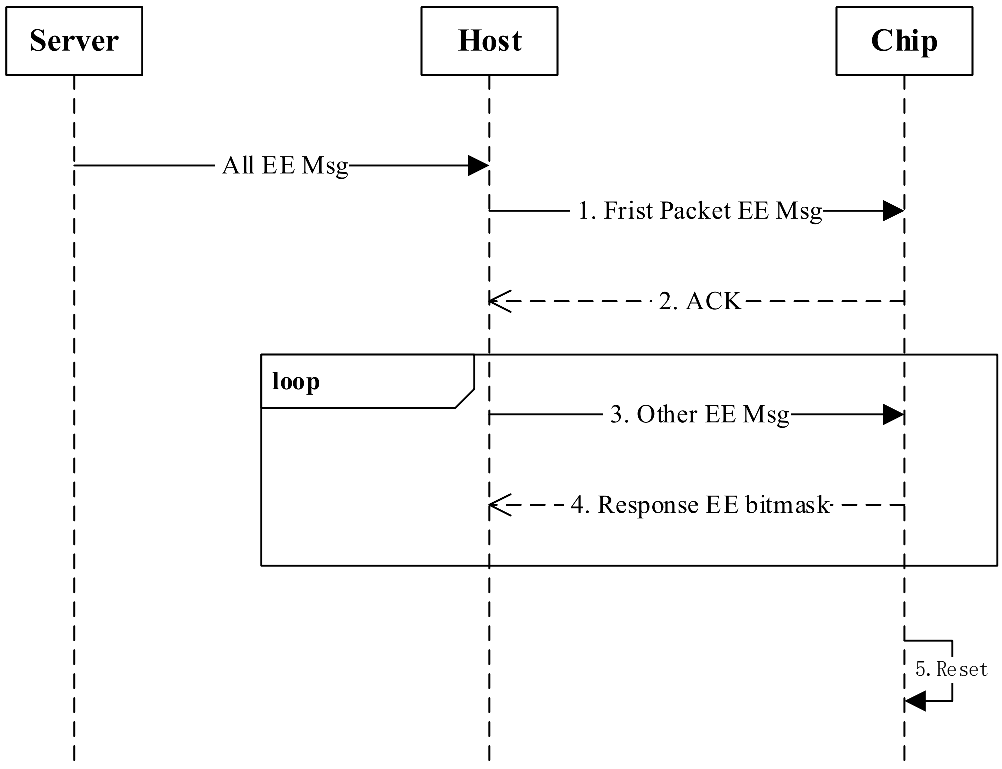
Each time the AGNSS ephemeris sent by the host is successfully received, the chip will respond with the current ephemeris collection status. Under normal operating conditions, this message can be ignored. You can pay attention to this message during confirmation and debugging. The message format is as follows:

Table 6-1 BK_AGNSS_Send_Ack_Respond

<table>
<tr><td colspan="2">Message</td><td colspan="6">BK_AGNSS_Send_Ack_Respond</td></tr>
<tr><td colspan="2">Description</td><td colspan="6">An ACK message contains a bitmask showing the satellite IDs of the received satellite ephemeris.</td></tr>
<tr><td colspan="2">Comment</td><td colspan="6">BK custom message type, send in little endian.
The satellite data of GPS, BDS, GALILEO, and GLONASS are sequentially packaged.
For example:
0x11: Bitmask of GPS indicating that the chip received GPS1 and GPS5 ephemeris.
0x81: Bitmask of BDS indicating that hat the chip received BDS1 and BDS8 ephemeris.</td></tr>
<tr><td colspan="2" rowspan="2">Message Structure</td><td>Header</td><td>MTYPE</td><td colspan="2">STYPE</td><td colspan="2">LEN (Bytes)</td></tr>
<tr><td>0x42 0x4B</td><td>0x1D</td><td colspan="2">0x0</td><td colspan="2">32</td></tr>
<tr><td colspan="8">Payload description:</td></tr>
<tr><td>Byte Offset</td><td>Number Format</td><td colspan="2">Scale</td><td>Name</td><td>Unit</td><td colspan="2">Description</td></tr>
<tr><td>0</td><td>uint64</td><td colspan="2">-</td><td>Bitmask</td><td>-</td><td colspan="2">Bitmask of GPS</td></tr>
<tr><td>8</td><td>uint64</td><td colspan="2">-</td><td>Bitmask</td><td>-</td><td colspan="2">Bitmask of BDS</td></tr>
<tr><td>16</td><td>uint64</td><td colspan="2">-</td><td>Bitmask</td><td>-</td><td colspan="2">Bitmask of GALILEO</td></tr>
<tr><td>24</td><td>uint64</td><td colspan="2">-</td><td>Bitmask</td><td>-</td><td colspan="2">Bitmask of GLONASS</td></tr>
</table>

6.7.2 Extended AGNSS Ephemeris Collection Situation

After the upper computer collects the extended ephemeris from the server, the specific process of injecting the extended ephemeris into the lower computer is shown in the following figure.



a) The Host sends the first packet of extended ephemeris Message. It is recommended that the Message size be set to at least 128 bytes to contain a complete customized RTCM message.
b) After decoding the extended ephemeris message, the lower computer (Chip) replies an ACK message to the upper computer;
c) After receiving the ACK message, the host computer can continue to inject the remaining extended ephemeris message;
d) The lower computer periodically restores the current extended ephemeris collection;
e) After the extended ephemeris is injected, the lower computer restarts automatically.

The format of the message that the lower machine replies ACK and ephemeris collection is as follows:

BK_EE_Send_Ack

Message	BK_EE_Send_Ack
Description	The lower machine replies Ack with the maximum injection rate supported
Comment	BK custom message type, send in little endian.

Message Structure	Header	MTYPE	STYPE	Length (Bytes)	
	0x42 0x4B	0x3C	0x0	2	
Payload description:					
Byte Offset	Number Format	Scale	Name	Unit	Description
0	UINT16	-	Inject rate	Byte/s	The maximum injection rate supported by the lower machine

BK_EE_Send_Response

Message	BK_EE_Send_Response				
Description	An ACK message contains a bitmask showing the satellite IDs of the received satellite ephemeris.				
Comment	BK custom message type, send in little endian. The satellite data of GPS, BDS, GLONASS and GALILEO and the four systems are sequentially packaged in Bitmask format, such as Bitmask of GPS: 0x11, indicating that GPS1 and 5 stars contain ephemeris information at this time.				
Message Structure	Header	MTYPE	STYPE	Length (Bytes)	
	0x42 0x4B	0x3C	0x1	32	
Payload description:					
Byte Offset	Number Format	Scale	Name	Unit	Description
0	UINT64	-	Bitmask	-	Bitmask of GPS
8	UINT64	-	Bitmask	-	Bitmask of BDS
16	UINT64	-	Bitmask	-	Bitmask of GLONASS
24	UINT64	-	Bitmask	-	Bitmask of GALILEO

6.8 Examples of Extended Ephemeris

The following is an example of the use of extended ephemeris using the upper computer software independently developed by BK

(1) Add an extended ephemeris related MQTT service to the configuration file

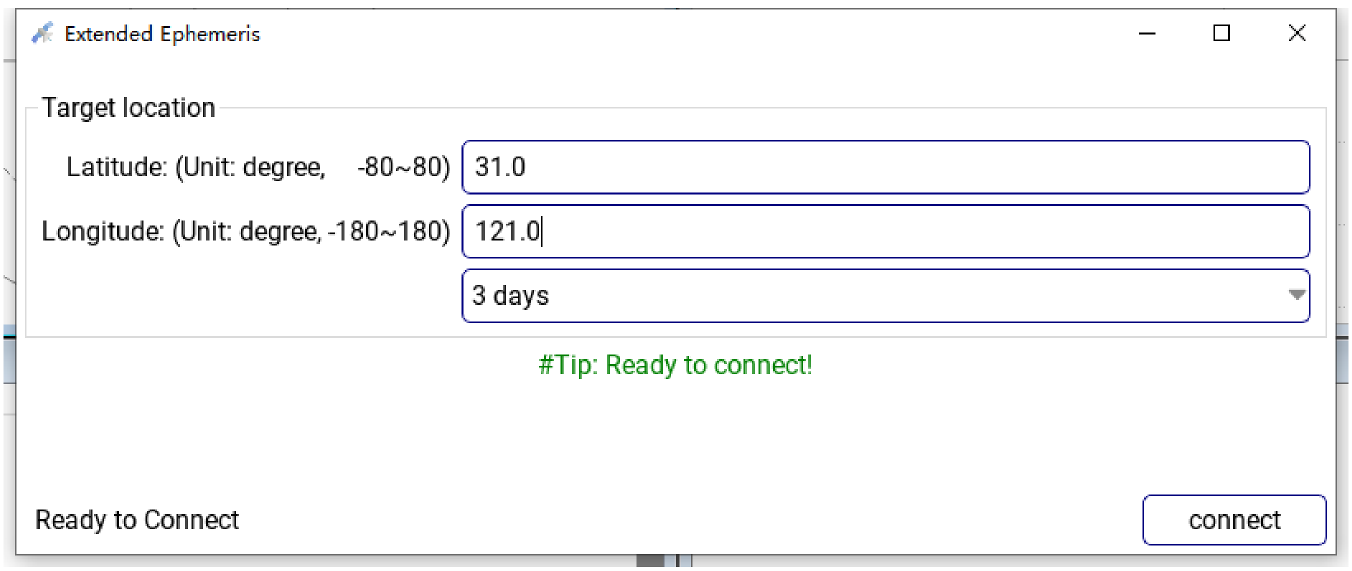
```
"Mqtt": {
    "SSL": true,
    "password": "[illegible]",
    "port": 1883,
    "server": "43.154.207.218",
    "user": "[illegible]",
    "valid": true
},
```

(2) Manually inject extended ephemeris

BK NavStar Tools supports extended ephemeris manual injection function

After the chip works normally, click "Port" and "Extend Ephemeris" on the menu bar to enter the longitude and latitude (such as Shanghai 31 N, 121 E), and select "connect" for forecast days.

After the extended ephemeris is injected, the receiver restarts automatically. After obtaining a valid local time, the receiver will load the corresponding extended ephemeris to attempt the first positioning.

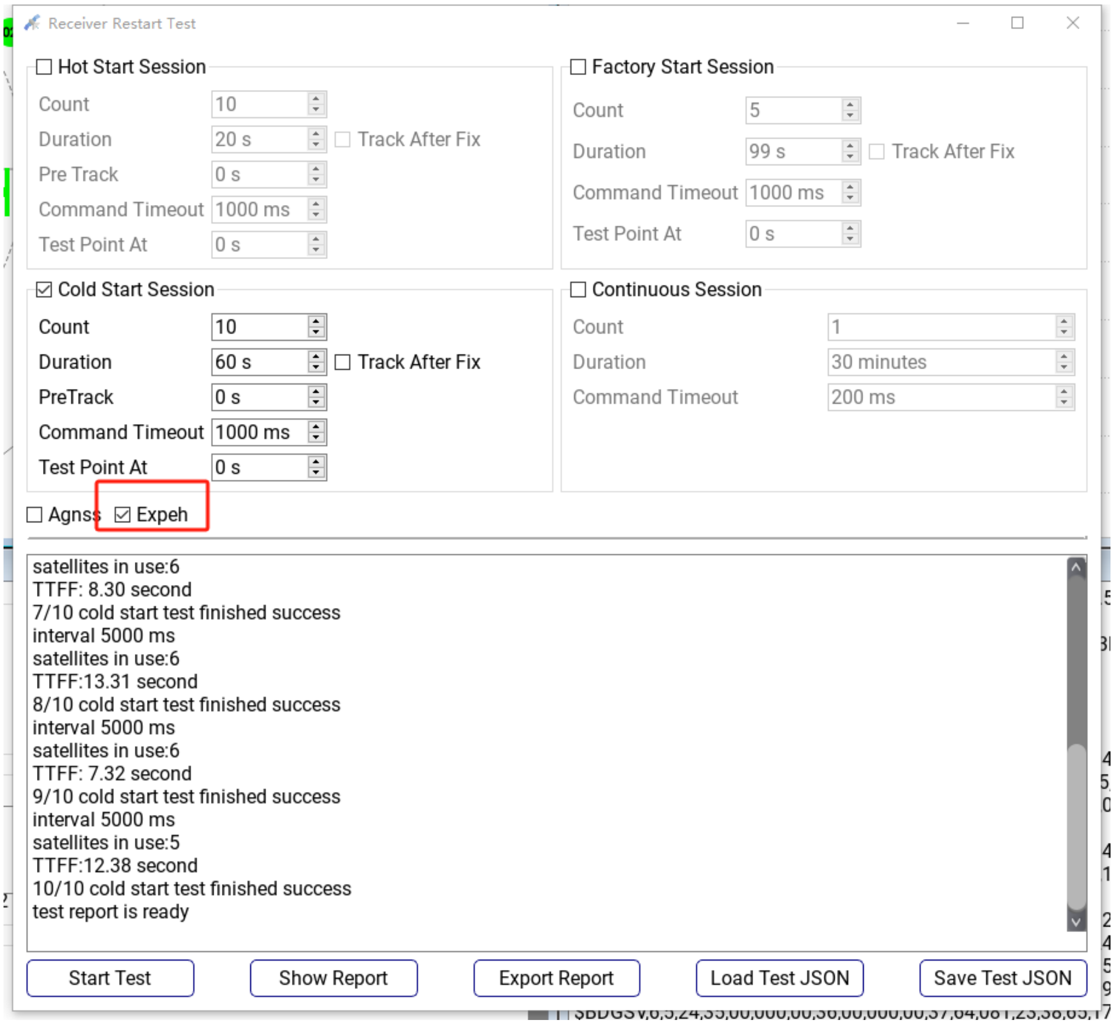


(3) Automatic injection of extended ephemeris

BK NavStar Tools supports extended ephemeris automatic injection and batch TTFF test functions.

After the next machine works normally, click "Receiver", "TTFF test", and select "Exeph" on the menu bar to perform automated test

Only the factory start test emptied the flash storage and automatically re-injected the extended ephemeris. hot/cold start does not automatically inject extended ephemeris. An extended ephemeris needs to be manually injected before batch positioning tests.



6.9 HTTP Subscription Ephemeris

6.9.1 Real-time Ephemeris (online)

The HTTP subscription URL is in the format of http://43.154.207.218 :8888/EPH +. The parameters can be in json format and can be submitted by post or directly in get form.

- POST Submit

{

user: user

password: password

sys:TGCER

lon:121

lat:31

}

- GET Submit

http://43.154.207.218 :8888/EPH/?user=user&password= password &sys= TGCER &lon=121&lat=31

Parameter Specification:

Parameter	Specification
user	Please contact the manufacturer to provide
password	Please contact the manufacturer to provide
sys	Indicates the type of subscription to ephemeris (in no particular alphabetical order) : T: Subscribe TIME. G: Subscribe to GPS ephemeris. C: Subscribes to BDS ephemeris; E: Subscribes to the Galileo ephemeris. R: Subscribes to the GLONASS ephemeris
lon	User approximate longitude, no need for precise position (unit: degree, range -180°~180°)
lat	User approximate latitude, no need for precise position (unit: degree, range -90°~90°)

6.9.2 Extending Ephemeris (offline)

The HTTP subscription URL is in the format of http://43.154.207.218 :8888/EEPH +. The parameters can be in json format and can be submitted by post or directly in get form.

- POST Submit

{

user: user

password: password

sys:TGCER

lon:121

lat:31

days:7

}

- GET Submit

http://43.154.207.218 :8888/EEPH/?user=user&password= password &sys= TGCER &lon=121&lat=31&days=7

Parameter Specification:

Parameter	Specification
user	Please contact the manufacturer to provide
password	Please contact the manufacturer to provide
sys	Indicates the type of subscription to ephemeris (in no particular alphabetical order) : T: Subscribe TIME. G: Subscribe to GPS ephemeris. C: Subscribes to BDS ephemeris; E: Subscribes to the Galileo ephemeris. R: Subscribes to the GLONASS ephemeris
lon	User approximate longitude, no need for precise position (unit: degree, range -180°~180°)
lat	User approximate latitude, no need for precise position (unit: degree, range -90°~90°)
days	Indicates the valid days of the subscription ephemeris, you can select 1\3\7, the default subscription is 7 days

6.9.3 HTTP Subscription Example

The following is an example of the BK166X series HTTP subscription AGNSS service and ephemeris injection. For ephemeris subscriptions, modify the corresponding ephemeris data for URL subscriptions as described in6.9.1 and 6.9.2.

```
#!/usr/bin/env python
# -*- encoding: utf-8 -*-
import socket
import time
```

```python
import serial

# ******[AGNSS Ephemeris Subscription]******* #
addr = "43.154.207.218"
port = 8888
path = '/EPH/?user=user&password=password&sys=TGCER&lon=121&lat=31'
request = f'GET {path} HTTP/1.1\r\nHost: {addr}\r\nConnection: close\r\n\r\n'
# request = f'POST {path} HTTP/1.1\r\nHost: {addr}\r\nConnection: close\r\n\r\n'

socket.setdefaulttimeout(4)
client = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
client.connect((addr, port))
client.send(request.encode("utf-8"))

replay_data = bytearray()
while True:
    cur_replay = client.recv(1024)
    if len(cur_replay) == 0:
        break
    else:
        replay_data.extend(cur_replay)
client.close()

# ******[BK166X AGNSS Test]******* #
ser = serial.Serial("com35", 115200)
time.sleep(1)
ser.write(b"$POLCFGRESET,1")  # BK166X Factory
time.sleep(1)

# Inject the Ephemeris BK166X
inject_len = 4*1024
cur_count = 0
cur_data = bytearray()
for tmp in replay_data:
    if cur_count >= inject_len:
        print("Current EPH injection length={}".format(len(cur_data)))
        ser.write(cur_data)
        cur_count = 0
        cur_data.clear()
    cur_data.append(tmp)
    cur_count += 1
```

```
if cur_count < inject_len:
    print("Current ephemeris injection length={}".format(len(cur_data)))
    ser.write(cur_data)

while True:
    time.sleep(1)
    data = ser.read_all()
    print(data.decode(encoding='gbk', errors='ignore'))
```

Revision History

Version	Date	Description
1.0	2022/4/25	Initial release.
1.1	2023/3/31	• Updated some descriptions and some commands. • Added Section 4 RTCM Protocol and Section 6 AGNSS. • Added NMEA THS message in Section 2.2 Standard Messages.
1.2	2023/4/17	• Added NMEA VTG message. • Added NMEA message output configuration command in Section 2.4 PO_NMEA Commands. • Added the description of version information. • Updated the description of the version number of the USRCFG commands in Section 3.4 Receiver Configuration Commands. • Added Section 5 Commonly Used Commands.
1.3	2023/4/25	Added dead reckoning solution message in Section 2.3 PO Information Messages.
1.4	2023/5/10	Updated Section 6 AGNSS.
1.5	2023/5/29	Added NMEA GLL and GST messages in Section 2.2 Standard Messages.
1.8	2023/9/6	• Updated Section 2.4 PO_NMEA Commands. • Updated Section 5 Commonly Used Commands. • Updated Section 6 AGNSS.
1.9	2023/9/19	Added IMU sensor raw data output message in Section 3.5 General Messages.
1.9.1	2023/11/21	Minor changes.
1.9.2	2023/11/22	Minor changes in Section 2.3 PO Information Messages .
1.9.3	2023/12/19	Added $PODUA sentence description.
1.9.4	2024/1/2	Added configure the base station coordinates Commands in Section 2.4
1.9.5	2024/1/8	Modify the GPS L1 L5 + IRS L5 PO commands
1.9.6	2024/1/9	Fixed B2A error described as B2C.
1.9.7	2024/2/4	Add description of NMEA ZDA message;
1.9.8	2024/3/7	Add the GPS L1 L5 + BDS B1I B2A B1C + GLO G1 + GAL E1 E5A + IRS L5 PO commands in Section 5.1.5

Version	Date	Description
1.9.9	2024/3/14	Fixed the description of the PO_ASCII command about configuring the NMEA rate
1.9.10	2024/3/26	Add AID, MSM, GMP, ODO, ATT, GFA,ZDA the PO_ASCII and Binary configuration commands
1.9.11	2024/3/29	Detailed description of the plaintext command configuration NMEA
1.9.12	2024/4/12	Add the description of attitude angles and driving behavior states in the NMEA_PO_INS statement
1.9.14	2024/5/14	Add 5.2.19 setting NMEA decimal digit
1.9.15	2024/5/21	Add description of carrier velocity and driving mileage and work time length in NMEA_PO_INS statement
1.9.16	2024/5/28	Revised history description simplified; Add 2.3.11 and modify 2.3.5 about version description; Add enable/disable SBASL1 signal hex cmd.
1.9.17	2024/6/28	Fixed hex closing and opening all NMEA commands; Added Extended Ephemeris HTTP subscription description; Added HTTP AGNSS test cases; Table 2-7, the signal ID of Beidou B2I is changed from "4" to "5", and the signal ID "4" is removed.
1.9.18	2024/7/15	Add 2.3.12 and 2.3.13;
1.9.19	2024/7/16	Add description of carrier acceleration in NMEA_PO_INS statement
1.9.20	2024/7/30	Added the description of PO_NMEA 2.4.11 Configuring PPS
1.9.21	2024/8/7	Added the description of 2.3.14 POTIM Message and 5.2.22 Setting Trigger Time Information
1.9.22	2024/9/4	Added 3.4 PNT message section, added 3.4.1 NAV message 2.4.6 Constellation Signal Configuration adds signal configuration mode Added Chip Information Inquiry Commands in Section 5.2.3
1.9.23	2024/9/9	Add description of carrier angular velocity in NMEA_PO_INS statement
1.9.24	2024/9/18	Add 2.4.13 Configure the Test Mode of Product Line Improved 3.6.1Deep Sleep Description.
1.9.25	2024/9/30	Add 2.2.11 NMEA_DTM,2.2.12 NMEA_GBS, 2.2.13 NMEA_GNS Output protocol description, and add 2.4.14 Configuring the Mincnr Value in PO_NMEA command.
1.9.26	2024/10/28	Added the description of 5.1.1 PO_NMEA command and 5.2.1 Binary Command reply.

© 2022-2024 Beken Corporation

Version	Date	Description
1.9.27	2024/10/29	Add the description of PO_NMEA configuration commands of integrated navigation: 2.4.15 Configure GNSS antenna lever arm of Integrated Navigation; 2.4.16 Configure Wheel Odometer Lever Arm of Integrated Navigation; 2.4.17 Configure IMU Mounted Axis of Integrated Navigation; 2.4.18 Configure IMU Mounted Misalignment of Integrated Navigation; 2.4.19 Configure Debug Log Output of Integrated Navigation.
1.9.28	2024/11/13	Add new PO_NMEA command for configuration output rate and order of a certain NMEA sentence type in 2.4.5.

Copyright

© 2023 Beken Corporation. The term "Beken" refers to Beken Corporation and/or its affiliates. This document contains information that is proprietary to Beken. Any unauthorized use, reproduction, or disclosure of this document in whole or in part is prohibited.

Disclaimer

The documentation is provided on an "as-is" basis only. Beken reserves the right to make any updates, corrections and any other modifications to its documentation without further notice and limitation to product information, descriptions, and specifications herein. Beken does not give warranties as to the accuracy or completeness of the included information. Beken shall have no liability for any use of the information in this documentation. You should obtain the latest relevant information before placing orders and should verify that such information is current and complete. Information published by Beken regarding any third-party products does not constitute a license to use such products or a warranty or endorsement thereof. Use of such information may require a license from a third party under the intellectual property rights of such third party, or a license from Beken under the intellectual property rights of Beken.

Trademarks

Beken, the BEKEN logo, and combinations thereof are trademarks or registered trademarks of Beken. All other product or brand names mentioned herein are trademarks or registered trademarks of their respective holders.

Beken Corporation

Building 41, 1387 Zhangdong Rd
Shanghai 201203
China

http://www.bekencorp.com